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Algebraic Riccati Equations

We have studied the Lyapunov equation in Chapter 3 and have seen the roles it played in some applications. A more general equation than the Lyapunov equation in control theory is the so-called *Algebraic Riccati Equation* or ARE for short. Roughly speaking, Lyapunov equations are most useful in system analysis while AREs are most useful in control system synthesis; particularly, they play the central roles in $\mathcal{H}_2$ and $\mathcal{H}_\infty$ optimal control.

Let A, Q, and R be real $n \times n$ matrices with Q and R symmetric. Then an algebraic Riccati equation is the following matrix equation:

$$A^*X + XA + XRX + Q = 0. \tag{13.1}$$

Associated with this Riccati equation is a $2n \times 2n$ matrix:

$$H := \begin{bmatrix} A & R \\ -Q & -A^* \end{bmatrix}. \tag{13.2}$$

A matrix of this form is called a *Hamiltonian matrix*. The matrix H in (13.2) will be used to obtain the solutions to the equation (13.1). It is useful to note that $\sigma(H)$ (the spectrum of H) is symmetric about the imaginary axis. To see that, introduce the $2n \times 2n$ matrix:

$$J := \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}$$

having the property $J^2 = -I$. Then

$$J^{-1}HJ = -JHJ = -H^*$$

so H and $-H^*$ are similar. Thus λ is an eigenvalue iff $-\bar{\lambda}$ is.

This chapter is devoted to the study of this algebraic Riccati equation and related problems: the properties of its solutions, the methods to obtain the solutions, and some applications.

In Section 13.1, we will study all solutions to (13.1). The word "all" means that any X, which is not necessarily real, not necessarily hermitian, not necessarily nonnegative, and not necessarily stabilizing, satisfies equation (13.1). The conditions for a solution to be hermitian, real, and so on, are also given in this section. The most important part of this chapter is Section 13.2 which focuses on the stabilizing solutions. This section is designed to be essentially self-contained so that readers who are only interested in the stabilizing solution may go to Section 13.2 directly without any difficulty. Section 13.3 presents the extreme (i.e., maximal or minimal) solutions of a Riccati equation and their properties. The relationship between the stabilizing solution of a Riccati equation and the spectral factorization of some frequency domain function is established in Section 13.4. Positive real functions and inner functions are introduced in Section 13.5 and 13.6. Some other special rational matrix factorizations, e.g., inner-outer factorizations and normalized coprime factorization, are given in Sections 13.7-13.8.

13.1 All Solutions of A Riccati Equation

The following theorem gives a way of constructing solutions to (13.1) in terms of invariant subspaces of H.

Theorem 13.1 *Let $\mathcal{V} \subset \mathbb{C}^{2n}$ be an n-dimensional invariant subspace of H, and let $X_1, X_2 \in \mathbb{C}^{n\times n}$ be two complex matrices such that*

$$\mathcal{V} = \operatorname{Im} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}.$$

If X_1 is invertible, then $X := X_2 X_1^{-1}$ is a solution to the Riccati equation (13.1) and $\sigma(A + RX) = \sigma(H|_{\mathcal{V}})$. Furthermore, the solution X is independent of a specific choice of bases of $\mathcal{V}$.

Proof. Since $\mathcal{V}$ is an H invariant subspace, there is a matrix $\Lambda \in \mathbb{C}^{n\times n}$ such that

$$\begin{bmatrix} A & R \\ -Q & -A^* \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \Lambda.$$

Postmultiply the above equation by X_1^{-1} to get

$$\begin{bmatrix} A & R \\ -Q & -A^* \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} X_1 \Lambda X_1^{-1}. \tag{13.3}$$

Now pre-multiply (13.3) by $\begin{bmatrix} -X & I \end{bmatrix}$ to get

$$\begin{aligned} 0 &= \begin{bmatrix} -X & I \end{bmatrix} \begin{bmatrix} A & R \\ -Q & -A^* \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} \\ &= -XA - A^*X - XRX - Q, \end{aligned}$$

which shows that X is indeed a solution of (13.1). Equation (13.3) also gives

$$A + RX = X_1 \Lambda X_1^{-1};$$

therefore, $\sigma(A+RX) = \sigma(\Lambda)$. But, by definition, Λ is a matrix representation of the map $H|_{\mathcal{V}}$, so $\sigma(A+RX) = \sigma(H|_{\mathcal{V}})$. Finally note that any other basis spanning $\mathcal{V}$ can be represented as

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} P = \begin{bmatrix} X_1 P \\ X_2 P \end{bmatrix}$$

for some nonsingular matrix P. The conclusion follows from the fact $(X_2P)(X_1P)^{-1} = X_2X_1^{-1}$. □

As we would expect, the converse of the theorem also holds.

Theorem 13.2 *If $X \in \mathbb{C}^{n\times n}$ is a solution to the Riccati equation (13.1), then there exist matrices $X_1, X_2 \in \mathbb{C}^{n\times n}$, with X_1 invertible, such that $X = X_2X_1^{-1}$ and the columns of $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ form a basis of an n-dimensional invariant subspace of H.*

Proof. Define $\Lambda := A + RX$. Multiplying this by X gives

$$X\Lambda = XA + XRX = -Q - A^*X$$

with the second equality coming from the fact that X is a solution to (13.1). Write these two relations as

$$\begin{bmatrix} A & R \\ -Q & -A^* \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} \Lambda.$$

Hence, the columns of $\begin{bmatrix} I \\ X \end{bmatrix}$ span an n-dimensional invariant subspace of H, and defining $X_1 := I$, and $X_2 := X$ completes the proof. □

Remark 13.1 It is now clear that to obtain solutions to the Riccati equation, it is necessary to be able to construct bases for those invariant subspaces of H. One way of constructing those invariant subspaces is to use eigenvectors and generalized eigenvectors of H. Suppose λ_i is an eigenvalue of H with multiplicity k (then $\lambda_{i+j} = \lambda_i$ for all $j = 1, \ldots, k-1$), and let v_i be a corresponding eigenvector and $v_{i+1}, \ldots, v_{i+k-1}$ be the corresponding generalized eigenvectors associated with v_i and λ_i. Then v_j are related by

$$\begin{aligned}
(H - \lambda_i I)v_i &= 0 \\
(H - \lambda_i I)v_{i+1} &= v_i \\
&\vdots \\
(H - \lambda_i I)v_{i+k-1} &= v_{i+k-2},
\end{aligned}$$

and the span$\{v_j, j = i, \ldots, i+k-1\}$ is an invariant subspace of H. ♡

Example 13.1 Let

$$A = \begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The eigenvalues of H are $1, 1, -1, -1$, and the corresponding eigenvector and generalized eigenvector of 1 are

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ -2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ -3/2 \\ 1 \\ 0 \end{bmatrix}.$$

The corresponding eigenvector and generalized eigenvector of -1 are

$$v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 1 \\ 3/2 \\ 0 \\ 0 \end{bmatrix}.$$

All solutions of the Riccati equation under various combinations are given below:

- span$\{v_1, v_2\}$ is H-invariant: let $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [v_1 \; v_2]$, then $X = X_2 X_1^{-1} = \begin{bmatrix} -10 & 6 \\ 6 & -4 \end{bmatrix}$ is a solution and $\sigma(A + RX) = \{1, 1\}$;
- span$\{v_1, v_3\}$ is H-invariant: let $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [v_1 \; v_3]$, then $X = X_2 X_1^{-1} = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$ is also a solution and $\sigma(A + RX) = \{1, -1\}$;

- $\text{span}\{v_3, v_4\}$ is H-invariant: let $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [v_3 \ v_4]$, then $X = 0$ is a solution and $\sigma(A + RX) = \{-1, -1\}$;
- $\text{span}\{v_1, v_4\}$, $\text{span}\{v_2, v_3\}$, and $\text{span}\{v_2, v_4\}$ are not H-invariant subspaces. Readers can verify that the matrices constructed from those vectors are not solutions to the Riccati equation (13.1).

$\diamond$

Up to this point, we have said nothing about the structure of the solutions given by Theorem 13.1 and 13.2. The following theorem gives a sufficient condition for a Riccati solution to be hermitian (not necessarily real symmetric).

Theorem 13.3 *Let $\mathcal{V}$ be an n-dimensional H-invariant subspace and let $X_1, X_2 \in \mathbb{C}^{n\times n}$ be such that*

$$\mathcal{V} = \text{Im} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}.$$

Then $\lambda_i + \bar{\lambda}_j \neq 0$ for all $i, j = 1, \ldots, n$, $\lambda_i, \lambda_j \in \sigma(H|_{\mathcal{V}})$ implies that $X_1^* X_2$ is hermitian, i.e., $X_1^* X_2 = (X_1^* X_2)^*$. Furthermore, if X_1 is nonsingular, then $X = X_2 X_1^{-1}$ is hermitian.

Proof. Since $\mathcal{V}$ is an invariant subspace of H, there is a matrix representation Λ for $H|_{\mathcal{V}}$ such that $\sigma(\Lambda) = \sigma(H|_{\mathcal{V}})$ and

$$H \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \Lambda.$$

Pre-multiply this equation by $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* J$ to get

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* JH \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* J \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \Lambda. \tag{13.4}$$

Note that JH is hermitian (actually symmetric since H is real); therefore, the left-hand side of (13.4) is hermitian as well as the right-hand side:

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* J \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \Lambda = \Lambda^* \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* J^* \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = -\Lambda^* \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* J \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

i.e.,

$$(-X_1^* X_2 + X_2^* X_1)\Lambda = -\Lambda^*(-X_1^* X_2 + X_2^* X_1).$$

This is a Lyapunov equation. Since $\lambda_i + \bar{\lambda}_j \neq 0$, the equation has a unique solution:

$$-X_1^* X_2 + X_2^* X_1 = 0.$$

This implies that $X_1^* X_2$ is hermitian.

That X is hermitian is easy to see by noting that $X = (X_1^{-1})^*(X_1^* X_2) X_1^{-1}$. □

Remark 13.2 It is clear from Example 13.1 that the condition $\lambda_i + \bar{\lambda}_j \neq 0$ is not necessary for the existence of a hermitian solution. ♡

The following theorem gives necessary and sufficient conditions for a solution to be real.

Theorem 13.4 *Let $\mathcal{V}$ be an n-dimensional H-invariant subspace, and let $X_1, X_2 \in \mathbb{C}^{n\times n}$ be such that X_1 is nonsingular and the columns of $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ form a basis of $\mathcal{V}$. Then $X := X_2 X_1^{-1}$ is real if and only if $\mathcal{V}$ is conjugate symmetric, i.e., $v \in \mathcal{V}$ implies that $\bar{v} \in \mathcal{V}$.*

Proof. ($\Leftarrow$) Since $\mathcal{V}$ is conjugate symmetric, there is a nonsingular matrix $P \in \mathbb{C}^{n\times n}$ such that

$$\begin{bmatrix} \bar{X}_1 \\ \bar{X}_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} P$$

where the over bar $\bar{X}$ denotes the complex conjugate. Therefore, $\bar{X} = \bar{X}_2 \bar{X}_1^{-1} = X_2 P (X_1 P)^{-1} = X_2 X_1^{-1} = X$ is real as desired.

($\Rightarrow$) Define $X := X_2 X_1^{-1}$. By assumption, $X \in \mathbb{R}^{n\times n}$ and

$$\operatorname{Im} \begin{bmatrix} I \\ X \end{bmatrix} = \mathcal{V};$$

therefore, $\mathcal{V}$ is conjugate symmetric. □

Example 13.2 This example is intended to show that there are non-real, non-hermitian solutions to equation (13.1). It is also designed to show that the condition $\lambda_i + \bar{\lambda}_j \neq 0, \forall\, i, j$, which excludes the possibility of having imaginary axis eigenvalues since if $\lambda_l = j\omega$ then $\lambda_l + \bar{\lambda}_l = 0$, is not necessary for the existence of a hermitian solution. Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & -4 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then H has eigenvalues $\lambda_1 = 1 = -\lambda_2, \lambda_3 = \lambda_5 = j = -\lambda_4 = -\lambda_6$. It is easy to show that

$$X = \begin{bmatrix} 0 & 0.5+0.5j & 0.5-0.5j \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

satisfies equation (13.1) and that X is neither real nor hermitian. On the other hand,

$$X = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is a real symmetric nonnegative definite solution corresponding to the eigenvalues -1, j, $-j$. ◇

13.2 Stabilizing Solution and Riccati Operator

In this section, we discuss when a solution is stabilizing, i.e., $\sigma(A+RX) \subset \mathbb{C}_-$ and the properties of such solutions. This section is the central part of this chapter and is designed to be self-contained. Hence some of the material appearing in this section may be similar to that seen in the previous section.

Assume H has no eigenvalues on the imaginary axis. Then it must have n eigenvalues in Re $s < 0$ and n in Re $s > 0$. Consider the two n-dimensional spectral subspaces, $\mathcal{X}_-(H)$ and $\mathcal{X}_+(H)$: the former is the invariant subspace corresponding to eigenvalues in Re $s < 0$ and the latter corresponds to eigenvalues in Re $s > 0$. By finding a basis for $\mathcal{X}_-(H)$, stacking the basis vectors up to form a matrix, and partitioning the matrix, we get

$$\mathcal{X}_-(H) = \mathrm{Im} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

where $X_1, X_2 \in \mathbb{C}^{n\times n}$. ($X_1$ and X_2 can be chosen to be real matrices.) If X_1 is nonsingular or, equivalently, if the two subspaces

$$\mathcal{X}_-(H), \quad \mathrm{Im} \begin{bmatrix} 0 \\ I \end{bmatrix} \tag{13.5}$$

are complementary, we can set $X := X_2 X_1^{-1}$. Then X is uniquely determined by H, i.e., $H \longmapsto X$ is a function, which will be denoted *Ric*. We will take the domain of *Ric*, denoted *dom*(*Ric*), to consist of Hamiltonian matrices H with two properties: H has no eigenvalues on the imaginary axis and the two subspaces in (13.5) are complementary. For ease of reference, these will be called the *stability property* and the *complementarity* *property*, respectively. This solution will be called the *stabilizing solution*. Thus, $X = Ric(H)$ and

$$Ric: \ dom(Ric) \subset \mathbb{R}^{2n\times 2n} \longmapsto \mathbb{R}^{n\times n}.$$

The following well-known results give some properties of X as well as verifiable conditions under which H belongs to $dom(Ric)$.

Theorem 13.5 *Suppose $H \in dom(Ric)$ and $X = Ric(H)$. Then*

(i) X is real symmetric;

(ii) X satisfies the algebraic Riccati equation

$$A^*X + XA + XRX + Q = 0;$$

(iii) $A + RX$ is stable .

Proof. (i) Let X_1, X_2 be as above. It is claimed that

$$X_1^*X_2 \text{ is symmetric.} \tag{13.6}$$

To prove this, note that there exists a stable matrix H_- in $\mathbb{R}^{n\times n}$ such that

$$H\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_- \quad .$$

(H_- is a matrix representation of $H|_{\mathcal{X}_-(H)}$.) Pre-multiply this equation by

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* J$$

to get

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* JH \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}^* J \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_- \quad . \tag{13.7}$$

Since JH is symmetric, so is the left-hand side of (13.7) and so is the right-hand side:

$$\begin{aligned}(-X_1^*X_2 + X_2^*X_1)H_- &= H_-^*(-X_1^*X_2 + X_2^*X_1)^* \\ &= -H_-^*(-X_1^*X_2 + X_2^*X_1).\end{aligned}$$

This is a Lyapunov equation. Since H_- is stable, the unique solution is

$$-X_1^*X_2 + X_2^*X_1 = 0.$$

This proves (13.6).

Since X_1 is nonsingular and $X = (X_1^{-1})^*(X_1^*X_2)X_1^{-1}$, X is symmetric.

(ii) Start with the equation

$$H\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_-$$

and post-multiply by X_1^{-1} to get

$$H\begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} X_1 H_- X_1^{-1}. \tag{13.8}$$

Now pre-multiply by $[X \quad -I]$:

$$[X \quad -I]H\begin{bmatrix} I \\ X \end{bmatrix} = 0.$$

This is precisely the Riccati equation.

(iii) Pre-multiply (13.8) by $[I \quad 0]$ to get

$$A + RX = X_1 H_- X_1^{-1}.$$

Thus $A + RX$ is stable because H_- is. □

Now, we are going to state one of the main theorems of this section which gives the necessary and sufficient conditions for the existence of a unique stabilizing solution of (13.1) under certain restrictions on the matrix R.

Theorem 13.6 *Suppose H has no imaginary eigenvalues and R is either positive semi-definite or negative semi-definite. Then $H \in dom(Ric)$ if and only if (A, R) is stabilizable.*

Proof. ($\Leftarrow$) To prove that $H \in dom(Ric)$, we must show that

$$\mathcal{X}_-(H), \quad \mathrm{Im}\begin{bmatrix} 0 \\ I \end{bmatrix}$$

are complementary. This requires a preliminary step. As in the proof of Theorem 13.5 define X_1, X_2, H_- so that

$$\mathcal{X}_-(H) = \mathrm{Im}\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$H\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} H_- \quad . \tag{13.9}$$

We want to show that X_1 is nonsingular, i.e., Ker $X_1 = 0$. First, it is claimed that Ker X_1 is H_--invariant. To prove this, let $x \in \operatorname{Ker} X_1$. Pre-multiply (13.9) by $[I \quad 0]$ to get

$$AX_1 + RX_2 = X_1 H_- \quad . \tag{13.10}$$

Pre-multiply by $x^* X_2^*$, post-multiply by x, and use the fact that $X_2^* X_1$ is symmetric (see (13.6)) to get

$$x^* X_2^* R X_2 x = 0.$$

Since R is semidefinite, this implies that $RX_2 x = 0$. Now post-multiply (13.10) by x to get $X_1 H_- x = 0$, i.e. $H_- x \in \operatorname{Ker} X_1$. This proves the claim.

Now to prove that X_1 is nonsingular, suppose, on the contrary, that Ker $X_1 \neq 0$. Then $H_-|_{\operatorname{Ker} X_1}$ has an eigenvalue, λ, and a corresponding eigenvector, x:

$$H_- x = \lambda x \tag{13.11}$$

$$\operatorname{Re} \lambda < 0, \quad 0 \neq x \in \operatorname{Ker} X_1.$$

Pre-multiply (13.9) by $[0 \quad I]$:

$$- QX_1 - A^* X_2 = X_2 H_- \quad . \tag{13.12}$$

Post-multiply the above equation by x and use (13.11):

$$(A^* + \lambda I) X_2 x = 0.$$

Recall that $RX_2 x = 0$, we have

$$x^* X_2^* [A + \overline{\lambda} I \quad R] = 0.$$

Then stabilizability of (A, R) implies $X_2 x = 0$. But if both $X_1 x = 0$ and $X_2 x = 0$, then $x = 0$ since $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ has full column rank, which is a contradiction.

($\Rightarrow$) This is obvious since $H \in dom(Ric)$ implies that X is a stabilizing solution and that $A + RX$ is asymptotically stable. It also implies that (A, R) must be stabilizable. $\square$

Theorem 13.7 *Suppose H has the form*

$$H = \begin{bmatrix} A & -BB^* \\ -C^*C & -A^* \end{bmatrix}.$$

Then $H \in dom(Ric)$ iff (A, B) is stabilizable and (C, A) has no unobservable modes on the imaginary axis. Furthermore, $X = Ric(H) \geq 0$ if $H \in dom(Ric)$, and $\operatorname{Ker}(X) = 0$ *if and only if (C, A) has no stable unobservable modes.*

Note that $\operatorname{Ker}(X) \subset \operatorname{Ker}(C)$, so that the equation $XM = C^*$ always has a solution for M, and a minimum F-norm solution is given by $X^\dagger C^*$.

Proof. It is clear from Theorem 13.6 that the stabilizability of (A, B) is necessary, and it is also sufficient if H has no eigenvalues on the imaginary axis. So we only need to show that, assuming (A, B) is stabilizable, H has no imaginary eigenvalues iff (C, A) has no unobservable modes on the imaginary axis. Suppose that $j\omega$ is an eigenvalue and $0 \neq \begin{bmatrix} x \\ z \end{bmatrix}$ is a corresponding eigenvector. Then

$$Ax - BB^*z = j\omega x$$

$$-C^*Cx - A^*z = j\omega z.$$

Re-arrange:

$$(A - j\omega I)x = BB^*z \tag{13.13}$$

$$-(A - j\omega I)^*z = C^*Cx. \tag{13.14}$$

Thus

$$\langle z, (A - j\omega I)x\rangle = \langle z, BB^*z\rangle = \|B^*z\|^2$$

$$-\langle x, (A - j\omega I)^*z\rangle = \langle x, C^*Cx\rangle = \|Cx\|^2$$

so $\langle x, (A - j\omega I)^*z\rangle$ is real and

$$-\|Cx\|^2 = \langle (A - j\omega I)x, z\rangle = \overline{\langle z, (A - j\omega I)x\rangle} = \|B^*z\|^2.$$

Therefore $B^*z = 0$ and $Cx = 0$. So from (13.13) and (13.14)

$$(A - j\omega I)x = 0$$

$$(A - j\omega I)^*z = 0.$$

Combine the last four equations to get

$$z^*[A - j\omega I \quad B] = 0$$

$$\begin{bmatrix} A - j\omega I \\ C \end{bmatrix} x = 0.$$

The stabilizability of (A, B) gives $z = 0$. Now it is clear that $j\omega$ is an eigenvalue of H iff $j\omega$ is an unobservable mode of (C, A).

Next, set $X := Ric(H)$. We'll show that $X \geq 0$. The Riccati equation is

$$A^*X + XA - XBB^*X + C^*C = 0$$

or equivalently

$$(A - BB^*X)^*X + X(A - BB^*X) + XBB^*X + C^*C = 0. \tag{13.15}$$

Noting that $A - BB^*X$ is stable (Theorem 13.5), we have

$$X = \int_0^\infty e^{(A-BB^*X)^*t}(XBB^*X + C^*C)e^{(A-BB^*X)t}dt. \tag{13.16}$$

Since $XBB^*X + C^*C$ is positive semi-definite, so is X.

Finally, we'll show that $\mathrm{Ker}X$ is non-trivial if and only if (C, A) has stable unobservable modes. Let $x \in \mathrm{Ker}X$, then $Xx = 0$. Pre-multiply (13.15) by x^* and post-multiply by x to get

$$Cx = 0.$$

Now post-multiply (13.15) again by x to get

$$XAx = 0.$$

We conclude that $\mathrm{Ker}(X)$ is an A-invariant subspace. Now if $\mathrm{Ker}(X) \neq 0$, then there is a $0 \neq x \in \mathrm{Ker}(X)$ and a λ such that $\lambda x = Ax = (A - BB^*X)x$ and $Cx = 0$. Since $(A - BB^*X)$ is stable, $\mathrm{Re}\lambda < 0$; thus λ is a stable unobservable mode. Conversely, suppose (C, A) has an unobservable stable mode λ, i.e., there is an x such that $Ax = \lambda x, Cx = 0$. By pre-multiplying the Riccati equation by x^* and post-multiplying by x, we get

$$2\mathrm{Re}\lambda x^*Xx - x^*XBB^*Xx = 0.$$

Hence $x^*Xx = 0$, i.e., X is singular. □

Example 13.3 This example shows that the observability of (C, A) is not necessary for the existence of a positive definite stabilizing solution. Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \; B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \; C = \begin{bmatrix} 0 & 0 \end{bmatrix}.$$

Then (A, B) is stabilizable, but (C, A) is not detectable. However,

$$X = \begin{bmatrix} 18 & -24 \\ -24 & 36 \end{bmatrix} > 0$$

is the stabilizing solution. ◇

Corollary 13.8 *Suppose that (A, B) is stabilizable and (C, A) is detectable. Then the Riccati equation*

$$A^*X + XA - XBB^*X + C^*C = 0$$

has a unique positive semidefinite solution. Moreover, the solution is stabilizing.

Proof. It is obvious from the above theorem that the Riccati equation has a unique stabilizing solution and that the solution is positive semidefinite. Hence we only need to show that any positive semidefinite solution $X \geq 0$ must also be stabilizing. Then by the uniqueness of the stabilizing solution, we can conclude that there is only one positive semidefinite solution. To achieve that goal, let us assume that $X \geq 0$ satisfies the Riccati equation but that it is not stabilizing. First rewrite the Riccati equation as

$$(A - BB^*X)^*X + X(A - BB^*X) + XBB^*X + C^*C = 0 \tag{13.17}$$

and let λ and x be an unstable eigenvalue and the corresponding eigenvector of $A - BB^*X$, respectively, i.e.,

$$(A - BB^*X)x = \lambda x.$$

Now pre-multiply and postmultiply equation (13.17) by x^* and x, respectively, and we have

$$(\bar{\lambda} + \lambda)x^*Xx + x^*(XBB^*X + CC)x = 0.$$

This implies

$$B^*Xx = 0, \quad Cx = 0$$

since $\mathrm{Re}(\lambda) \geq 0$ and $X \geq 0$. Finally, we arrive at

$$Ax = \lambda x, \quad Cx = 0$$

i.e., (C, A) is not detectable, which is a contradiction. Hence $\mathrm{Re}(\lambda) < 0$, i.e., $X \geq 0$ is the stabilizing solution. □

Lemma 13.9 *Suppose D has full column rank and let $R = D^*D > 0$; then the following statements are equivalent:*

(i) $\begin{bmatrix} A - j\omega I & B \\ C & D \end{bmatrix}$ *has full column rank for all ω.*

(ii) $\big((I - DR^{-1}D^*)C,\ A - BR^{-1}D^*C\big)$ *has no unobservable modes on $j\omega$-axis.*

Proof. Suppose $j\omega$ is an unobservable mode of $\big((I - DR^{-1}D^*)C,\ A - BR^{-1}D^*C\big)$; then there is an $x \neq 0$ such that

$$(A - BR^{-1}D^*C)x = j\omega x, \quad (I - DR^{-1}D^*)Cx = 0$$

i.e.,

$$\begin{bmatrix} A - j\omega I & B \\ C & D \end{bmatrix} \begin{bmatrix} I & 0 \\ -R^{-1}D^*C & I \end{bmatrix} \begin{bmatrix} x \\ 0 \end{bmatrix} = 0.$$

But this implies that
$$\begin{bmatrix} A - j\omega I & B \\ C & D \end{bmatrix} \tag{13.18}$$
does not have full column rank. Conversely, suppose (13.18) does not have full column rank for some ω; then there exists $\begin{bmatrix} u \\ v \end{bmatrix} \neq 0$ such that
$$\begin{bmatrix} A - j\omega I & B \\ C & D \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 0.$$
Now let
$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} I & 0 \\ -R^{-1}D^*C & I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$
Then
$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} I & 0 \\ R^{-1}D^*C & I \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} \neq 0$$
and
$$(A - BR^{-1}D^*C - j\omega I)x + By = 0 \tag{13.19}$$
$$(I - DR^{-1}D^*)Cx + Dy = 0. \tag{13.20}$$
Pre-multiply (13.20) by D^* to get $y = 0$. Then we have
$$(A - BR^{-1}D^*C)x = j\omega x, \quad (I - DR^{-1}D^*)Cx = 0$$
i.e., $j\omega$ is an unobservable mode of $\big((I - DR^{-1}D^*)C,\, A - BR^{-1}D^*C\big)$. □

Remark 13.3 If D is not square, then there is a $D_\perp$ such that $\begin{bmatrix} D_\perp & DR^{-1/2} \end{bmatrix}$ is unitary and that $D_\perp D_\perp^* = I - DR^{-1}D^*$. Hence, in some cases we will write the condition (ii) in the above lemma as $(D_\perp^* C, A - BR^{-1}D^*C)$ having no imaginary unobservable modes. Of course, if D is square, the condition is simplified to $A - BR^{-1}D^*C$ with no imaginary eigenvalues. Note also that if $D^*C = 0$, condition (ii) becomes (C, A) with no imaginary unobservable modes. ♡

Corollary 13.10 *Suppose D has full column rank and denote $R = D^*D > 0$. Let H have the form*
$$\begin{aligned} H &= \begin{bmatrix} A & 0 \\ -C^*C & -A^* \end{bmatrix} - \begin{bmatrix} B \\ -C^*D \end{bmatrix} R^{-1} \begin{bmatrix} D^*C & B^* \end{bmatrix} \\ &= \begin{bmatrix} A - BR^{-1}D^*C & -BR^{-1}B^* \\ -C^*(I - DR^{-1}D^*)C & -(A - BR^{-1}D^*C)^* \end{bmatrix}. \end{aligned}$$

Then $H \in dom(Ric)$ iff (A, B) is stabilizable and $\begin{bmatrix} A - j\omega I & B \\ C & D \end{bmatrix}$ *has full column rank for all ω. Furthermore, $X = Ric(H) \geq 0$ if $H \in dom(Ric)$, and $Ker(X) = 0$ if and only if $(D_\perp^* C, A - BR^{-1}D^*C)$ has no stable unobservable modes.*

Proof. This is the consequence of the Lemma 13.9 and Theorem 13.7. □

Remark 13.4 It is easy to see that the detectability (observability) of $(D_\perp^* C, A - BR^{-1}D^*C)$ implies the detectability (observability) of (C, A); however, the converse is in general not true. Hence the existence of a stabilizing solution to the Riccati equation in the above corollary is not guaranteed by the stabilizability of (A, B) and detectability of (C, A). Furthermore, even if a stabilizing solution exists, the positive definiteness of the solution is not guaranteed by the observability of (C, A) unless $D^*C = 0$. As an example, consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \; C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \; D = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Then (C, A) is observable, (A, B) is controllable, and

$$A - BD^*C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad D_\perp^* C = \begin{bmatrix} 0 & 0 \end{bmatrix}.$$

A Riccati equation with the above data has a nonnegative definite stabilizing solution since $(D_\perp^* C, A - BR^{-1}D^*C)$ has no unobservable modes on the imaginary axis. However, the solution is not positive definite since $(D_\perp^* C, A - BR^{-1}D^*C)$ has a stable unobservable mode. On the other hand, if the B matrix is changed to

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

then the corresponding Riccati equation has no stabilizing solution since, in this case, $(A - BD^*C)$ has eigenvalues on the imaginary axis although (A, B) is controllable and (C, A) is observable. ♡

13.3 Extreme Solutions and Matrix Inequalities

We have shown in the previous sections that given a Riccati equation, there are generally many possible solutions. Among all solutions, we are most interested in those which are real, symmetric, and, in particular, the stabilizing solutions. There is another class of solutions which are interesting; they are called *extreme* (maximal or minimal) solutions. Some properties of the extreme solutions will be studied in this section. The connections between the extreme solutions and the stabilizing solutions will also be established in this section. To illustrate the idea, let us look at an example.

Example 13.4 Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \ B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \ C = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}.$$

The eigenvalues of matrix $H = \begin{bmatrix} A & -BB^* \\ -C^*C & -A^* \end{bmatrix}$ are

$$\lambda_1 = 1, \ \lambda_2 = -1, \ \lambda_3 = 2, \ \lambda_4 = -2, \ \lambda_5 = -3, \ \lambda_6 = 3,$$

and their corresponding eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 2 \\ -3 \\ 6 \\ 0 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, v_4 = \begin{bmatrix} 4 \\ 3 \\ -12 \\ 0 \\ 12 \\ 0 \end{bmatrix}, v_5 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, v_6 = \begin{bmatrix} -3 \\ -6 \\ -1 \\ 0 \\ 0 \\ 6 \end{bmatrix}.$$

There are four distinct nonnegative definite symmetric solutions depending on the chosen invariant subspaces:

- $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [v_1 \ v_3 \ v_5]; \ Y_1 = X_2 X_1^{-1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix};$
- $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [v_2 \ v_3 \ v_5]; \ Y_2 = X_2 X_1^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix};$
- $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [v_1 \ v_4 \ v_5]; \ Y_3 = X_2 X_1^{-1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix};$
- $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = [v_2 \ v_4 \ v_5]; \ Y_4 = X_2 X_1^{-1} = \begin{bmatrix} 18 & -24 & 0 \\ -24 & 36 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$

These solutions can be ordered as $Y_4 \geq Y_i \geq Y_1$, $i = 2, 3$. Of course, this is only a partial ordering since Y_2 and Y_3 are not comparable. Note also that only Y_4 is a stabilizing solution, i.e., $A - BB^*Y_4$ is stable. Furthermore, Y_4 and Y_1 are the "maximal" and "minimal" solutions, respectively. ◇

The partial ordering concept shown in Example 13.4 can be stated in a much more general setting. To do that, again consider the Riccati equation (13.1). We shall call a hermitian solution X_+ of (13.1) a *maximal solution* if $X_+ \geq X$ for all hermitian solutions X of (13.1). Similarly, we shall call a hermitian solution X_- of (13.1) a *minimal solution* if $X_- \leq X$ for all hermitian solutions X of (13.1). Clearly, maximal and minimal solutions are unique if they exist.

To study the properties of the maximal and minimal solutions, we shall introduce the following quadratic matrix:

$$\mathcal{Q}(X) := A^*X + XA + XRX + Q. \tag{13.21}$$

Theorem 13.11 *Assume $R \leq 0$ and assume there is a hermitian matrix $X = X^*$ such that $\mathcal{Q}(X) \geq 0$.*

(i) If (A, R) is stabilizable, then there exists a unique maximal solution X_+ to the Riccati equation (13.1). Furthermore,

$$X_+ \geq X, \ \forall X \text{ such that } \mathcal{Q}(X) \geq 0$$

and $\sigma(A + RX_+) \subset \overline{\mathbb{C}}_-$.

(ii) If $(-A, R)$ is stabilizable, then there exists a unique minimal solution X_- to the Riccati equation (13.1). Furthermore,

$$X_- \leq X, \ \forall X \text{ such that } \mathcal{Q}(X) \geq 0$$

and $\sigma(A + RX_-) \subset \overline{\mathbb{C}}_+$.

(iii) If (A, R) is controllable, then both X_+ and X_- exist. Furthermore, $X_+ > X_-$ iff $\sigma(A + RX_+) \subset \mathbb{C}_-$ iff $\sigma(A + RX_-) \subset \mathbb{C}_+$. In this case,

$$\begin{aligned} X_+ - X_- &= \left(\int_0^\infty e^{(A+RX_+)t} R e^{(A+RX_+)^* t} dt \right)^{-1} \\ &= \left(\int_{-\infty}^0 e^{(A+RX_-)t} R e^{(A+RX_-)^* t} dt \right)^{-1}. \end{aligned}$$

(iv) If $\mathcal{Q}(X) > 0$, the results in (i) and (ii) can be respectively strengthened to $X_+ > X$, $\sigma(A + RX_+) \subset \mathbb{C}_-$, and $X_- < X$, $\sigma(A + RX_-) \subset \mathbb{C}_+$.

Proof. Let $R = -BB^*$ for some B. Note the fact that (A, R) is stabilizable (controllable) iff (A, B) is.

(i): Let X be such that $\mathcal{Q}(X) \geq 0$. Since (A, B) is stabilizable, there is an F_0 such that

$$A_0 := A + BF_0$$

is stable. Now let X_0 be the unique solution to the Lyapunov equation

$$X_0 A_0 + A_0^* X_0 + F_0^* F_0 + Q = 0.$$

Then X_0 is hermitian. Define

$$\hat{F}_0 := F_0 + B^* X,$$

and we have the following equation:

$$(X_0 - X)A_0 + A_0^*(X_0 - X) = -\hat{F}_0^* \hat{F}_0 - \mathcal{Q}(X) \leq 0.$$

The stability of A_0 implies that

$$X_0 \geq X.$$

Starting with X_0, we shall define a non-increasing sequence of hermitian matrices $\{X_i\}$. Associated with $\{X_i\}$, we shall also define a sequence of stable matrices $\{A_i\}$ and a sequence of matrices $\{F_i\}$. Assume inductively that we have already defined matrices $\{X_i\}$, $\{A_i\}$, and $\{F_i\}$ for i up to $n-1$ such that X_i is hermitian and

$$X_0 \geq X_1 \geq \cdots \geq X_{n-1} \geq X,$$

$$A_i = A + BF_i, \text{ is stable}, \ i = 0, \ldots, n-1;$$

$$F_i = -B^* X_{i-1}, \ i = 1, \ldots, n-1;$$

$$X_i A_i + A_i^* X_i = -F_i^* F_i - Q, \ i = 0, 1, \ldots, n-1. \tag{13.22}$$

Next, introduce

$$F_n = -B^* X_{n-1},$$

$$A_n = A + BF_n.$$

First we show that A_n is stable. Then, using (13.22), with $i = n$, we define a hermitian matrix X_n with $X_{n-1} \geq X_n \geq X$. Now using (13.22), with $i = n-1$, we get

$$X_{n-1} A_n + A_n^* X_{n-1} + Q + F_n^* F_n + (F_n - F_{n-1})^*(F_n - F_{n-1}) = 0. \tag{13.23}$$

Let

$$\hat{F}_n := F_n + B^* X;$$

then

$$(X_{n-1} - X)A_n + A_n^*(X_{n-1} - X) = -\mathcal{Q}(X) - \hat{F}_n^* \hat{F}_n - (F_n - F_{n-1})^*(F_n - F_{n-1}). \tag{13.24}$$

Now assume that A_n is not stable, i.e., there exists an λ with $\mathrm{Re}\lambda \geq 0$ and $x \neq 0$ such that $A_n x = \lambda x$. Then pre-multiply (13.24) by x^* and postmultiply by x, and we have

$$2\mathrm{Re}\lambda x^*(X_{n-1} - X)x = -x^*\{\mathcal{Q}(X) + \hat{F}_n^* \hat{F}_n + (F_n - F_{n-1})^*(F_n - F_{n-1})\}x.$$

Since it is assumed $X_{n-1} \geq X$, each term on the right-hand side of the above equation has to be zero. So we have

$$x^*(F_n - F_{n-1})^*(F_n - F_{n-1})x = 0.$$

This implies

$$(F_n - F_{n-1})x = 0.$$

But now

$$A_{n-1}x = (A + BF_{n-1})x = (A + BF_n)x = A_n x = \lambda x,$$

which is a contradiction with the stability of A_{n-1}. Hence A_n is stable as well.

Now we introduce X_n as the unique solution of the Lyapunov equation

$$X_n A_n + A^* X_n = -F_n^* F_n - Q. \tag{13.25}$$

Then X_n is hermitian. Next, we have

$$(X_n - X)A_n + A_n^*(X_n - X) = -\mathcal{Q}(X) - \hat{F}_n^* \hat{F}_n \leq 0,$$

and, by using (13.23),

$$(X_{n-1} - X_n)A_n + A_n^*(X_{n-1} - X_n) = -(F_n - F_{n-1})^*(F_n - F_{n-1}) \leq 0.$$

Since A_n is stable, we have

$$X_{n-1} \geq X_n \geq X.$$

We have a non-increasing sequence $\{X_i\}$, and the sequence is bounded below by $X_i \geq X$. Hence the limit

$$X_f := \lim_{n\to\infty} X_n$$

exists and is hermitian, and we have $X_f \geq X$. Passing the limit $n \to \infty$ in (13.25), we get $\mathcal{Q}(X_f) = 0$. So X_f is a solution of (13.1). Since X is an arbitrary element satisfying $\mathcal{Q}(X) \geq 0$ and X_f is independent of the choice of X, we have

$$X_f \geq X, \ \forall \ X \text{ such that } \mathcal{Q}(X) \geq 0.$$

In particular, X_f is the maximal solution of the Riccati equation (13.1), i.e., $X_f = X_+$.

To establish the stability property of the maximal solution, note that A_n is stable for any n. Hence, in the limit, the eigenvalues of

$$A - BB^* X_f$$

will have non-positive real parts. The uniqueness follows from the fact that the maximal solution is unique.

(ii): The results follow by the following substitutions in the proof of part (i):

$$A \leftarrow -A; \ X \leftarrow -X; \ X_+ \leftarrow -X_-.$$

(iii): The existence of X_+ and X_- follows from (i) and (ii). Let $A_+ := A + RX_+$; we now show that $X_+ - X_- > 0$ iff $\sigma(A_+) \subset \mathbb{C}_-$. It is easy to verify that

$$A_+^*(X_+ - X_-) + (X_+ - X_-)A_+ - (X_+ - X_-)R(X_+ - X_-) = 0.$$

($\Rightarrow$): Suppose $X_+ - X_- > 0$; then $(A_+, R(X_+ - X_-))$ is controllable. Therefore, from Lyapunov theorem, we conclude that A_+ is stable.

($\Leftarrow$): Assuming now that A_+ is stable and that X is any other solution to the Riccati equation (13.1),

$$A_+^*(X_+ - X) + (X_+ - X)A_+ - (X_+ - X)R(X_+ - X) = 0.$$

This equation has an invertible solution

$$X_+ - \bar{X} = \left(- \int_0^\infty e^{(A+RX_+)t} R e^{(A+RX_+)^* t} dt \right)^{-1} > 0.$$

A simple rearrangement of terms gives

$$(A + R\bar{X})^*(X_+ - \bar{X}) + (X_+ - \bar{X})(A + R\bar{X}) + (X_+ - \bar{X})R(X_+ - \bar{X}) = 0.$$

Since $X_+ - \bar{X} > 0$, we conclude $\sigma(A + R\bar{X}) \subset \mathbb{C}_+$. This in turn implies $\bar{X} = X_-$. Therefore, $X_+ > X_-$. That $X_+ - X_- > 0$ iff $\sigma(A + RX_-) \subset \mathbb{C}_+$ follows by analogy.

(iv): We shall only show the case for X_+; the case for X_- follows by analogy. Note that from (i) we have

$$(X_+ - X)A_+ + A_+^*(X_+ - X) = -\mathcal{Q}(X) + (X_+ - X)R(X_+ - X) < 0 \qquad (13.26)$$

and $X_+ - X \geq 0$. Now suppose $X_+ - X$ is singular and there is an $x \neq 0$ such that $(X_+ - X)x = 0$. By pre-multiplying (13.26) by x^* and post-multiplying by x, we get $x^* \mathcal{Q}(X) x = 0$, a contradiction. The stability of $A + RX_+$ then follows from the Lyapunov theorem. □

Remark 13.5 The proof given above also gives an iterative procedure to compute the maximal and minimal solutions. For example, to find the maximal solution, the following procedures can be used:

(i) find F_0 such that $A_0 = A + BF_0$ is stable;

(ii) solve X_i: $X_i A_i + A_i^* X_i + F_i^* F_i + Q = 0$;

(iii) if $\|X_i - X_{i-1}\| \leq \epsilon$ =specified accuracy, stop. Otherwise go to (iv);

(iv) let $F_{i+1} = -B^* X_i$ and $A_{i+1} = A + BF_{i+1}$ go to (ii).

This procedure will converge to the stabilizing solution if the solution exists. ♡

Corollary 13.12 *Let $R \leq 0$ and suppose (A, R) is controllable and X_1, X_2 are two solutions to the Riccati equation (13.1). Then $X_1 > X_2$ implies that $X_+ = X_1$, $X_- = X_2$, $\sigma(A + RX_1) \subset \mathbb{C}_-$ and that $\sigma(A + RX_2) \subset \mathbb{C}_+$.*

The following example illustrates that the stabilizability of (A, R) is not sufficient to guarantee the existence of a minimal hermitian solution of equation (13.1).

Example 13.5 Let

$$A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \ R = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}, \ Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then it can be shown that all the hermitian solutions of (13.1) are given by

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \ \begin{bmatrix} -1 & \alpha \\ \bar{\alpha} & -\frac{1}{2}|\alpha|^2 \end{bmatrix}, \ \alpha \in \mathbb{C}.$$

The maximal solution is clearly

$$X_+ = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix};$$

however, there is no minimal solution. $\diamond$

The Riccati equation appeared in $\mathcal{H}_\infty$ control, which will be considered in the later part of this book, often has $R \geq 0$. However, these conditions are only a dual of the above theorem.

Corollary 13.13 *Assume that $R \geq 0$ and that there is a hermitian matrix $X = X^*$ such that $\mathcal{Q}(X) \leq 0$.*

(i) If (A, R) is stabilizable, then there exists a unique minimal solution X_- to the Riccati equation (13.1). Furthermore,

$$X_- \leq X, \ \forall X \text{ such that } \mathcal{Q}(X) \leq 0$$

and $\sigma(A + RX_-) \subset \overline{\mathbb{C}}_-$.

(ii) If $(-A, R)$ is stabilizable, then there exists a unique maximal solution X_+ to the Riccati equation (13.1). Furthermore,

$$X_+ \geq X, \ \forall X \text{ such that } \mathcal{Q}(X) \leq 0$$

and $\sigma(A + RX_+) \subset \overline{\mathbb{C}}_+$.

(iii) If (A,R) is controllable, then both X_+ and X_- exist. Furthermore, $X_+ > X_-$ iff $\sigma(A+RX_-) \subset \mathbb{C}_-$ iff $\sigma(A+RX_+) \subset \mathbb{C}_+$. In this case,

$$\begin{aligned} X_+ - X_- &= \left(\int_0^\infty e^{(A+RX_-)t} R e^{(A+RX_-)^* t} dt\right)^{-1} \\ &= \left(\int_{-\infty}^0 e^{(A+RX_+)t} R e^{(A+RX_+)^* t} dt\right)^{-1}. \end{aligned}$$

(iv) If $\mathcal{Q}(X) < 0$, the results in (i) and (ii) can be respectively strengthened to $X_+ > X$, $\sigma(A+RX_+) \subset \mathbb{C}_+$, and $X_- < X$, $\sigma(A+RX_-) \subset \mathbb{C}_-$.

Proof. The proof is similar to the proof for Theorem 13.11. □

Theorem 13.11 can be used to derive some comparative results for some Riccati equations. More specifically, let

$$H_s := \begin{bmatrix} A - BR_s^{-1}S^* & -BR_s^{-1}B^* \\ -P + SR_s^{-1}S^* & -(A - BR_s^{-1}S^*)^* \end{bmatrix}$$

and

$$\tilde{H}_s := \begin{bmatrix} \tilde{A} - \tilde{B}\tilde{R}_s^{-1}\tilde{S}^* & -\tilde{B}\tilde{R}_s^{-1}\tilde{B}^* \\ -\tilde{P} + \tilde{S}\tilde{R}_s^{-1}\tilde{S}^* & -(\tilde{A} - \tilde{B}\tilde{R}_s^{-1}\tilde{S}^*)^* \end{bmatrix}$$

where $P, \tilde{P}, R_s$, and $\tilde{R}_s$ are real symmetric and $R_s > 0$, $\tilde{R}_s > 0$. We shall also make use of the following matrices:

$$T := \begin{bmatrix} P & S \\ S^* & R_s \end{bmatrix}, \ \tilde{T} := \begin{bmatrix} \tilde{P} & \tilde{S} \\ \tilde{S}^* & \tilde{R}_s \end{bmatrix}.$$

We denote by X_+ and $\tilde{X}_+$ the maximal solution to the Riccati equation associated with H_s and $\tilde{H}_s$, respectively:

$$(A - BR_s^{-1}S^*)^*X + X(A - BR_s^{-1}S^*) - XBR_s^{-1}B^*X + (P - SR_s^{-1}S^*) = 0 \quad (13.27)$$

$$(\tilde{A} - \tilde{B}\tilde{R}_s^{-1}\tilde{S}^*)^*\tilde{X} + \tilde{X}(\tilde{A} - \tilde{B}\tilde{R}_s^{-1}\tilde{S}^*) - \tilde{X}\tilde{B}\tilde{R}_s^{-1}\tilde{B}^*\tilde{X} + (\tilde{P} - \tilde{S}\tilde{R}_s^{-1}\tilde{S}^*) = 0. \quad (13.28)$$

Recall that JH_s and $J\tilde{H}_s$ are hermitian where J is defined as

$$J = \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}.$$

Let

$$K := JH_s = \begin{bmatrix} P - SR_s^{-1}S^* & (A - BR_s^{-1}S^*)^* \\ A - BR_s^{-1}S^* & -BR_s^{-1}B^* \end{bmatrix}$$

$$\tilde{K} := J\tilde{H}_s = \begin{bmatrix} \tilde{P} - \tilde{S}\tilde{R}_s^{-1}\tilde{S}^* & (\tilde{A} - \tilde{B}\tilde{R}_s^{-1}\tilde{S}^*)^* \\ \tilde{A} - \tilde{B}\tilde{R}_s^{-1}\tilde{S}^* & -\tilde{B}\tilde{R}_s^{-1}\tilde{B}^* \end{bmatrix}.$$

Theorem 13.14 *Suppose* (A, B) *and* $(\tilde{A}, \tilde{B})$ *are stabilizable.*

(i) Assume that (13.28) has a hermitian solution and that $K \geq \tilde{K}$ $(K > \tilde{K})$*; then* X_+ *and* $\tilde{X}_+$ *exist, and* $X_+ \geq \tilde{X}_+$ $(X_+ > \tilde{X}_+)$.

(ii) Let $A = \tilde{A}$, $B = \tilde{B}$, *and* $T \geq \tilde{T}$ $(T > \tilde{T})$*. Assume that (13.28) has a hermitian solution. Then* X_+ *and* $\tilde{X}_+$ *exist, and* $X_+ \geq \tilde{X}_+$ $(X_+ > \tilde{X}_+)$.

(iii) If $T \geq 0$ $(T > 0)$*, then* X_+ *exists, and* $X_+ \geq 0$ $(X_+ > 0)$.

Proof. (i): Let X be a hermitian solution of (13.28); then we have

$$\begin{bmatrix} I \\ X \end{bmatrix}^* \tilde{K} \begin{bmatrix} I \\ X \end{bmatrix} = 0.$$

Then, since $K \geq \tilde{K}$, we have

$$\mathcal{Q}_s(X) := \begin{bmatrix} I \\ X \end{bmatrix}^* K \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix}^* (K - \tilde{K}) \begin{bmatrix} I \\ X \end{bmatrix} \geq 0.$$

Now we use Theorem 13.11 to obtain the existence of X_+, and $X_+ \geq X$. Since X is arbitrary, we get $X_+ \geq X$ for all hermitian solutions X of (13.28). The existence of $\tilde{X}_+$ follows immediately by applying Theorem 13.11. Moreover,

$$X_+ \geq \tilde{X}_+.$$

(ii): Let $A = \tilde{A}$, $B = \tilde{B}$ and let X be a hermitian solution of (13.28). Denote $L = -R_s^{-1}(S^* + B^*X)$ and $\tilde{L} = -\tilde{R}_s^{-1}(\tilde{S}^* + B^*X)$. Then

$$\mathcal{Q}_s(X) = \begin{bmatrix} I \\ X \end{bmatrix}^* K \begin{bmatrix} I \\ X \end{bmatrix} = A^*X + XA - L^*R_sL + P$$

while (13.28) becomes

$$A^*X + XA - \tilde{L}^*\tilde{R}_s\tilde{L} + \tilde{P} = 0.$$

It is easy to show that

$$\begin{aligned} \mathcal{Q}_s(X) &= \begin{bmatrix} I \\ X \end{bmatrix}^* (K - \tilde{K}) \begin{bmatrix} I \\ X \end{bmatrix} \\ &= P - \tilde{P} - L^*R_sL + \tilde{L}^*\tilde{R}_s\tilde{L} \\ &= \begin{bmatrix} I \\ L \end{bmatrix}^* (T - \tilde{T}) \begin{bmatrix} I \\ L \end{bmatrix} + (L - \tilde{L})^*\tilde{R}_s(L - \tilde{L}) \geq 0. \end{aligned}$$

Now as in part (i), there exist X_+ and $\tilde{X}_+$, and $X_+ \geq \tilde{X}_+$.

(iii): The condition $T \geq 0$ implies $P - SR_s^{-1}S^* \geq 0$, so we have $\mathcal{Q}_s(0) = P - SR_s^{-1}S^* \geq 0$. Apply Theorem 13.11 to get the existence of X_+, and $X_+ \geq 0$. □

13.4 Spectral Factorizations

Let A, B, P, S, R be real matrices of compatible dimensions such that $P = P^*$, $R = R^*$, and define a parahermitian rational matrix function

$$\begin{aligned}\Phi(s) &= R + S^*(sI-A)^{-1}B + B^*(-sI-A^*)^{-1}S + B^*(-sI-A^*)^{-1}P(sI-A)^{-1}B \\ &= \begin{bmatrix} B^*(-sI-A^*)^{-1} & I \end{bmatrix} \begin{bmatrix} P & S \\ S^* & R \end{bmatrix} \begin{bmatrix} (sI-A)^{-1}B \\ I \end{bmatrix}. \end{aligned} \tag{13.29}$$

Lemma 13.15 *Suppose R is nonsingular and either one of the following assumptions is satisfied:*

(A1) A has no eigenvalues on $j\omega$-axis;

(A2) P is sign definite, i.e., $P \geq 0$ or $P \leq 0$, (A, B) has no uncontrollable modes on $j\omega$-axis, and (P, A) has no unobservable modes on the $j\omega$-axis.

Then the following statements are equivalent:

(i) $\Phi(j\omega_0)$ is singular for some $0 \leq \omega_0 \leq \infty$.

(ii) The Hamiltonian matrix

$$H = \begin{bmatrix} A - BR^{-1}S^* & -BR^{-1}B^* \\ -(P - SR^{-1}S^*) & -(A - BR^{-1}S^*)^* \end{bmatrix}$$

has an eigenvalue at $j\omega_0$.

Proof. $(i) \Rightarrow (ii)$: Let

$$\Phi(s) = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array}\right] := \left[\begin{array}{cc|c} A & 0 & B \\ -P & -A^* & -S \\ \hline S^* & B^* & R \end{array}\right].$$

Then $H = \hat{A} - \hat{B}\hat{D}^{-1}\hat{C}$ and

$$\Phi^{-1}(s) = \left[\begin{array}{c|c} H & \begin{bmatrix} -BR^{-1} \\ SR^{-1} \end{bmatrix} \\ \hline \begin{bmatrix} R^{-1}S^* & R^{-1}B^* \end{bmatrix} & R^{-1} \end{array}\right].$$

If $\Phi(j\omega_0)$ is singular, then $j\omega_0$ is a zero of $\Phi(s)$. Hence $j\omega_0$ is a pole of $\Phi^{-1}(s)$, and $j\omega_0$ is an eigenvalue of H.

$(ii) \Rightarrow (i)$: Suppose $j\omega_0$ is an eigenvalue of H but is not a pole of $\Phi^{-1}(s)$. Then $j\omega_0$ must be either an unobservable mode of $(\begin{bmatrix} R^{-1}S^* & R^{-1}B^* \end{bmatrix}, H)$ or an uncontrollable mode of $(H, \begin{bmatrix} -BR^{-1} \\ SR^{-1} \end{bmatrix})$. Now suppose $j\omega_0$ is an unobservable mode of $(\begin{bmatrix} R^{-1}S^* & R^{-1}B^* \end{bmatrix}, H)$. Then there exists an $x_0 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq 0$ such that

$$Hx_0 = j\omega_0 x_0, \quad \begin{bmatrix} R^{-1}S^* & R^{-1}B^* \end{bmatrix} x_0 = 0.$$

These equations can be simplified to

$$(j\omega_0 I - A)x_1 = 0 \tag{13.30}$$

$$(j\omega_0 I + A^*)x_2 = -Px_1 \tag{13.31}$$

$$S^* x_1 + B^* x_2 = 0. \tag{13.32}$$

We now consider two cases under the different assumptions:

(a) If assumption (A1) is true, then $x_1 = 0$ from (13.30), and this, in turn, implies $x_2 = 0$ from (13.31), which is a contradiction.

(b) If assumption (A2) is true, since (13.30) implies $x_1^*(j\omega_0 I + A^*) = 0$, from (13.31) we have $x_1^* P x_1 = 0$. This gives $Px_1 = 0$ since P is sign definite ($P \geq 0$ or $P \leq 0$). This implies, along with (13.30) that $j\omega_0$ is an unobservable mode of (P, A) if $x_1 \neq 0$. On the other hand, if $x_1 = 0$, then (13.31) and (13.32) imply that (A, B) has an uncontrollable mode at $j\omega_0$, again a contradiction.

Similarly, a contradiction will also be derived if $j\omega_0$ is assumed to be an uncontrollable mode of $(H, \begin{bmatrix} -BR^{-1} \\ SR^{-1} \end{bmatrix})$. □

Corollary 13.16 *Suppose $R > 0$ and either one of the assumptions (A1) or (A2) defined in Lemma 13.15 is true. Then the following statements are equivalent:*

(i) $\Phi(j\omega) > 0$ *for all* $0 \leq \omega \leq \infty$.

(ii) The Hamiltonian matrix

$$H = \begin{bmatrix} A - BR^{-1}S^* & -BR^{-1}B^* \\ -(P - SR^{-1}S^*) & -(A - BR^{-1}S^*)^* \end{bmatrix}$$

has no eigenvalue on $j\omega$-axis.

Proof. $(i) \Rightarrow (ii)$ follows easily from Lemma 13.15. To prove $(ii) \Rightarrow (i)$, note that $\Phi(j\infty) = R > 0$ and $\det \Phi(j\omega) \neq 0$ for all ω from Lemma 13.15. Then the continuity of $\Phi(j\omega)$ gives $\Phi(j\omega) > 0$. $\square$

Lemma 13.17 *Suppose A is stable and $P \leq 0$. Then*

(i) the matrix

$$\Phi_0(s) = \begin{bmatrix} B^*(-sI - A^*)^{-1} & I \end{bmatrix} \begin{bmatrix} P & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} (sI - A)^{-1}B \\ I \end{bmatrix}$$

satisfies

$$\Phi_0(j\omega) > 0, \quad \text{for all } 0 \leq \omega \leq \infty$$

if and only if there exists a unique real $X = X^* \leq 0$ such that

$$A^*X + XA - XBB^*X + P = 0$$

and $\sigma(A - BB^*X) \subset \mathbb{C}_-$.

(ii) $\Phi_0(j\omega) \geq 0$, for all $0 \leq \omega \leq \infty$ if and only if there exists a unique real $X = X^* \leq 0$ such that

$$A^*X + XA - XBB^*X + P = 0 \tag{13.33}$$

and $\sigma(A - BB^*X) \subset \overline{\mathbb{C}}_-$.

Proof. (i): $(\Rightarrow)$ From Corollary 13.16, $\Phi_0(j\omega) > 0$ implies that a Hamiltonian matrix

$$\begin{bmatrix} A & -BB^* \\ -P & -A^* \end{bmatrix}$$

has no eigenvalues on the imaginary axis. This in turn implies from Theorem 13.6 that (13.33) has a stabilizing solution. Furthermore, since

$$\begin{bmatrix} A & -BB^* \\ -P & -A^* \end{bmatrix} = \begin{bmatrix} -I & \\ & I \end{bmatrix} \begin{bmatrix} A & BB^* \\ P & -A^* \end{bmatrix} \begin{bmatrix} -I & \\ & I \end{bmatrix},$$

we have

$$Ric \begin{bmatrix} A & -BB^* \\ -P & -A^* \end{bmatrix} = -Ric \begin{bmatrix} A & BB^* \\ P & -A^* \end{bmatrix} \leq 0.$$

$(\Leftarrow)$ The sufficiency proof is omitted here and is a special case of $(b) \Rightarrow (a)$ of Theorem 13.19 below.

(ii): $(\Rightarrow)$ Let $P = -C^*C$ and $G(s) := C(sI - A)^{-1}B$. Then $\Phi_0(s) = I - G^\sim(s)G(s)$. Let $0 < \gamma < 1$ and define

$$\Psi_\gamma(s) := I - \gamma^2 G^\sim(s)G(s).$$

Then $\Psi_\gamma(j\omega) > 0, \quad \forall\omega$. Thus from part (i), there is an $X_\gamma = X_\gamma^* \le 0$ such that $A - BB^*X_\gamma$ is stable and

$$A^*X_\gamma + X_\gamma A - X_\gamma BB^*X_\gamma - \gamma^2 C^*C = 0.$$

It is easy to see from Theorem 13.14 that X_γ is monotone-decreasing with γ, i.e., $X_{\gamma_1} \ge X_{\gamma_2}$ if $\gamma_1 \le \gamma_2$. To show that $\lim_{\gamma\to 1} X_\gamma$ exists, we need to show that X_γ is bounded below for all $0 < \gamma < 1$.

In the following, it will be assumed that (A, B) is controllable. The controllability assumption will be removed later.

Let Y be the destabilizing solution to the following Riccati equation:

$$A^*Y + YA - YBB^*Y = 0$$

with $\sigma(A - BB^*Y) \subset \mathbb{C}_+$ (note that the existence of such a solution is guaranteed by the controllability assumption). Then it is easy to verify that

$$(A-BB^*Y)^*(X_\gamma - Y) + (X_\gamma - Y)(A - BB^*Y) - (X_\gamma - Y)BB^*(X_\gamma - Y) - \gamma^2 C^*C = 0.$$

This implies that

$$X_\gamma - Y = \int_0^\infty e^{-(A-BB^*Y)^*t}[(X_\gamma - Y)BB^*(X_\gamma - Y) + \gamma^2 C^*C]e^{-(A-BB^*Y)t}dt \ge 0.$$

Thus X_γ is bounded below with Y as the lower bound, and $\lim_{\gamma\to 1} X_\gamma$ exists. Let $X := \lim_{\gamma\to 1} X_\gamma$; then from continuity argument, X satisfies the Riccati equation

$$A^*X + XA - XBB^*X + P = 0$$

and $\sigma(A - BB^*X) \subset \overline{\mathbb{C}}_-$.

Now suppose (A, B) is not controllable, and then assume without loss of generality that

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} C_1 & C_2 \end{bmatrix}$$

so that (A_{11}, B_1) is controllable and A_{11} and A_{22} are stable. Then the Riccati equation for

$$X_\gamma = \begin{bmatrix} X_{11}^\gamma & X_{12}^\gamma \\ (X_{12}^\gamma)^* & X_{22}^\gamma \end{bmatrix}$$

can be written as three equations

$$A_{11}^*X_{11}^\gamma + X_{11}^\gamma A_{11} - X_{11}^\gamma B_1B_1^*X_{11}^\gamma - \gamma^2 C_1^*C_1 = 0$$

$$(A_{11} - B_1B_1X_{11}^\gamma)^*X_{12}^\gamma + X_{12}^\gamma A_{22} + X_{11}^\gamma A_{12} - \gamma^2 C_1^*C_2 = 0$$

$$A_{22}^*X_{22}^\gamma + X_{22}^\gamma A_{22} + (X_{12}^\gamma)^*A_{12} + A_{12}^*X_{12}^\gamma - (X_{12}^\gamma)^*B_1B_1^*X_{12}^\gamma - \gamma^2 C_2^*C_2 = 0$$

and

$$A - BB^*X = \begin{bmatrix} A_{11} - B_1B_1^*X_{11}^\gamma & A_{12} - B_1B_1^*X_{12}^\gamma \\ 0 & A_{22} \end{bmatrix}$$

is stable.

Let Y_{11} be the anti-stabilizing solution to

$$A_{11}^*Y_{11} + Y_{11}A_{11} - Y_{11}B_1B_1^*Y_{11} = 0$$

with $\sigma(A_{11} - B_1B_1^*Y_{11}) \subset \mathbb{C}_+$. Then it is clear that $X_{11}^\gamma - Y_{11} \geq 0$. Moreover, $X_{11} = \lim_{\gamma\to 1} X_{11}^\gamma$ exists and satisfies the following Riccati equation:

$$A_{11}^*X_{11} + X_{11}A_{11} - X_{11}B_1B_1^*X_{11} - C_1^*C_1 = 0$$

with $\sigma(A_{11} - B_1B_1^*X_{11}) \subset \overline{\mathbb{C}}_-$. Consequently, the following Sylvester equation

$$(A_{11} - B_1B_1X_{11})^*X_{12} + X_{12}A_{22} + X_{11}A_{12} - C_1^*C_2 = 0$$

has a unique solution X_{12} since $\lambda_i(A_{11} - B_1B_1X_{11}) + \lambda_j(A_{22}) \neq 0,\ \forall i, j$. Furthermore, the following Lyapunov equation has a unique nonnegative definite solution X_{22}:

$$A_{22}^*X_{22} + X_{22}A_{22} + X_{12}^*A_{12} + A_{12}^*X_{12} - X_{12}^*B_1B_1^*X_{12} - C_2^*C_2 = 0.$$

We have proven that there exists a unique X such that

$$A^*X + XA - XBB^*X - C^*C = 0$$

and $\sigma(A - BB^*X) \subset \overline{\mathbb{C}}_-$.

($\Leftarrow$) same as in part (i). $\square$

Lemma 13.18 *Let $R > 0$,*

$$\Phi(s) = \begin{bmatrix} B^*(-sI - A^*)^{-1} & I \end{bmatrix} \begin{bmatrix} P & S \\ S^* & R \end{bmatrix} \begin{bmatrix} (sI - A)^{-1}B \\ I \end{bmatrix}$$

and

$$\hat{\Phi}(s) = \begin{bmatrix} \hat{B}^*(-sI - \hat{A}^*)^{-1} & I \end{bmatrix} \begin{bmatrix} \hat{P} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} (sI - \hat{A})^{-1}\hat{B} \\ I \end{bmatrix}$$

where

$$\hat{A} := A - BR^{-1}S^*,\ \hat{B} := BR^{-1/2},\ \hat{P} := P - SR^{-1}S^*.$$

Then $\Phi(j\omega) \geq 0$ iff $\hat{\Phi}(j\omega) \geq 0$.

Proof. Note that

$$\begin{bmatrix} P & S \\ S^* & R \end{bmatrix} = \begin{bmatrix} I & SR^{-1/2} \\ 0 & R^{1/2} \end{bmatrix} \begin{bmatrix} \hat{P} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} I & 0 \\ R^{-1/2}S^* & R^{1/2} \end{bmatrix}.$$

Hence the function $\Phi(s)$ can be written as

$$\Phi(s) = \begin{bmatrix} B^*(-sI - A^*)^{-1} & \varphi^{\sim}(s) \end{bmatrix} \begin{bmatrix} \hat{P} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} (sI - A)^{-1}B \\ \varphi(s) \end{bmatrix}$$

with $\varphi(s) = R^{1/2} + R^{-1/2}S^*(sI - A)^{-1}B$. It is easy to verify that

$$\hat{\Phi}(s) = [\varphi^{-1}(s)]^{\sim}\Phi(s)\varphi^{-1}(s).$$

Hence the conclusion follows. □

Now we are ready to state and prove one of the main results of this section. The following theorem and corollary characterize the relations among spectral factorizations, Riccati equations, and decomposition of Hamiltonians.

Theorem 13.19 *Let A, B, P, S, R be matrices of compatible dimensions such that $P = P^*$, $R = R^* > 0$, with (A, B) stabilizable. Suppose either one of the following assumptions is satisfied:*

(A1) A has no eigenvalues on $j\omega$-axis;

(A2) P is sign definite, i.e., $P \geq 0$ or $P \leq 0$ and (P, A) has no unobservable modes on the $j\omega$-axis.

Then

(I) The following statements are equivalent:

(a) The parahermitian rational matrix

$$\Phi(s) = \begin{bmatrix} B^*(-sI - A^*)^{-1} & I \end{bmatrix} \begin{bmatrix} P & S \\ S^* & R \end{bmatrix} \begin{bmatrix} (sI - A)^{-1}B \\ I \end{bmatrix}$$

satisfies

$$\Phi(j\omega) > 0 \text{ for all } 0 \leq \omega \leq \infty.$$

(b) There exists a unique real $X = X^*$ such that

$$(A - BR^{-1}S^*)^*X + X(A - BR^{-1}S^*) - XBR^{-1}B^*X + P - SR^{-1}S^* = 0$$

and that $A - BR^{-1}S^* - BR^{-1}B^*X$ is stable.

(c) The Hamiltonian matrix

$$H = \begin{bmatrix} A - BR^{-1}S^* & -BR^{-1}B^* \\ -(P - SR^{-1}S^*) & -(A - BR^{-1}S^*)^* \end{bmatrix}$$

has no $j\omega$-axis eigenvalues.

(II) The following statements are also equivalent:

(d) $\Phi(j\omega) \geq 0$ for all $0 \leq \omega \leq \infty$.

(e) There exists a unique real $X = X^*$ such that

$$(A - BR^{-1}S^*)^*X + X(A - BR^{-1}S^*) - XBR^{-1}B^*X + P - SR^{-1}S^* = 0$$

and that $\sigma(A - BR^{-1}S^* - BR^{-1}B^*X) \subset \overline{\mathbb{C}}_-$.

The following corollary is usually referred to as the spectral factorization theory.

Corollary 13.20 *If either one of the conditions, (a)–(e), in Theorem 13.19 is satisfied, then there exists an $M \in \mathcal{R}_p$ such that*

$$\Phi = M^\sim RM.$$

A particular realization of one such M is

$$M = \left[\begin{array}{c|c} A & B \\ \hline -F & I \end{array}\right]$$

where $F = -R^{-1}(S^* + B^*X)$. Furthermore, if X is the stabilizing solution, then $M^{-1} \in \mathcal{RH}_\infty$.

Remark 13.6 If the stabilizability of (A, B) is changed into the stabilizability of $(-A, B)$, then the theorem still holds except that the solutions X in (b) and (e) are changed into the destabilizing solution ($\sigma(A - BR^{-1}S^* - BR^{-1}B^*X) \subset \mathbb{C}_+$) and the weakly destabilizing solution ($\sigma(A - BR^{-1}S^* - BR^{-1}B^*X) \subset \overline{\mathbb{C}}_+$), respectively. ♡

Proof. $(a) \Rightarrow (c)$ follows from Corollary 13.16.

$(c) \Rightarrow (b)$ follows from Theorem 13.6 and Theorem 13.5.

$(b) \Rightarrow (a)$ Suppose $\exists X = X^*$ such that $A - BR^{-1}S^* - BR^{-1}B^*X = A - BR^{-1}(S^* + B^*X)$ is stable. Let $F = -R^{-1}(S^* + B^*X)$ and

$$M = \left[\begin{array}{c|c} A & B \\ \hline -F & I \end{array}\right].$$

It is easily verified by use of the Riccati equation for X and by routine algebra that $\Phi = M^{\sim}RM$. Since

$$M^{-1} = \left[\begin{array}{c|c} A+BF & B \\ \hline F & I \end{array}\right],$$

$M^{-1} \in \mathcal{RH}_\infty$. Thus $M(s)$ has no zeros on the imaginary axis and $\Phi(j\omega) > 0$.

$(e) \Rightarrow (d)$ follows the same procedure as the proof of $(b) \Rightarrow (a)$.

$(d) \Rightarrow (e)$: Assume $S = 0$ and $R = I$; otherwise use Lemma 13.18 first to get a new function with such properties. Let $P = C_1^*C_1 - C_2^*C_2$ with C_1 and C_2 square nonsingular. Note that this decomposition always exists since $P = \alpha I - (\alpha I - P)$, with $\alpha > 0$ sufficiently large, defines one such possibility. Let X_1 be the positive definite solution to

$$A^*X_1 + X_1A - X_1BB^*X_1 + C_1^*C_1 = 0.$$

By Theorem 13.7, X_1 indeed exists and is stabilizing, i.e., $A_1 := A - BB^*X_1$ is stable. Let $\Delta = X - X_1$. Then the equation in Δ becomes

$$A_1^*\Delta + \Delta A_1 - \Delta BB^*\Delta - C_2^*C_2 = 0.$$

To show that this equation has a solution, recall Lemma 13.17 and note that A_1 is stable; then it is sufficient to show that

$$I - B^*(-j\omega I - A_1^*)^{-1}C_2^*C_2(j\omega I - A_1)^{-1}B$$

is positive semi-definite.

Notice first that

$$C_2(sI - A + BB^*X_1)^{-1}B = C_2(sI-A)^{-1}B\left[I + B^*X_1(sI-A)^{-1}B\right]^{-1}.$$

From the definition of X_1 and Corollary 13.20, we also have that

$$\begin{aligned} &I + B^*(-sI - A^*)^{-1}C_1^*C_1(sI-A)^{-1}B \\ &= \left[I + B^*X_1(sI-A)^{-1}B\right]^{\sim}\left[I + B^*X_1(sI-A)^{-1}B\right]. \end{aligned}$$

Now, by assumption

$$\begin{aligned} \Phi(j\omega) = I &+ B^*(-j\omega I - A^*)^{-1}C_1^*C_1(j\omega I - A)^{-1}B \\ &- B^*(-j\omega I - A^*)^{-1}C_2^*C_2(j\omega I - A)^{-1}B \ge 0. \end{aligned}$$

It follows that

$$\begin{aligned} I - &\left\{\left[I + B^*(-j\omega I - A^*)^{-1}X_1B\right]^{-1}B^*(-j\omega I - A^*)^{-1}C_2^*\right\} \\ &\left\{C_2(j\omega I - A)^{-1}B\left[I + B^*X_1(j\omega I - A)^{-1}B\right]^{-1}\right\} \ge 0. \end{aligned}$$

Consequently,

$$I - B^*(-j\omega I - A_1^*)^{-1}C_2^*C_2(j\omega I - A_1)^{-1}B \geq 0.$$

We may now apply Lemma 13.17 to the Δ equation. Consequently, there exists a unique solution Δ such that $\sigma(A_1 - BB^*\Delta) = \sigma(A - BB^*X) \subset \overline{\mathbb{C}}_-$. This shows the existence and uniqueness of X.

□

We shall now illustrate the proceeding results through a simple example. Note in particular that the function can have poles on the imaginary axis.

Example 13.6 Let $A = 0, B = 1, R = 1, S = 0$ and $P = 1$. Then $\Phi(s) = 1 - \frac{1}{s^2}$ and $\Phi(j\omega) > 0$. It is easy to check that the Hamiltonian matrix

$$H = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

does not have eigenvalues on the imaginary axis and $X = 1$ is the stabilizing solution to the corresponding Riccati equation and the spectral factor is given by

$$M(s) = \left[\begin{array}{c|c} 0 & 1 \\ \hline 1 & 1 \end{array}\right] = \frac{s+1}{s}.$$

◇

Some frequently used special spectral factorizations are now considered.

Corollary 13.21 *Assume that* $G(s) := \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{RL}_\infty$ *is a stabilizable and detectable realization and* $\gamma > \|G(s)\|_\infty$. *Then, there exists a transfer matrix* $M \in \mathcal{RL}_\infty$ *such that* $M^\sim M = \gamma^2 I - G^\sim G$ *and* $M^{-1} \in \mathcal{RH}_\infty$. *A particular realization of* M *is*

$$M(s) = \left[\begin{array}{c|c} A & B \\ \hline -R^{1/2}F & R^{1/2} \end{array}\right]$$

where

$$\begin{aligned} R &= \gamma^2 I - D^*D \\ F &= R^{-1}(B^*X + D^*C) \\ X &= Ric\begin{bmatrix} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix} \end{aligned}$$

and $X \geq 0$ *if* A *is stable.*

Proof. This is a special case of Theorem 13.19. In fact, the theorem follows by letting $P = -C^*C$, $S = -C^*D$, $R = \gamma^2 I - D^*D$ in Theorem 13.19 and by using the fact that

$$\mathrm{Ric}\begin{bmatrix} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix} =$$

$$-\mathrm{Ric}\begin{bmatrix} A + BR^{-1}D^*C & -BR^{-1}B^* \\ C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix}.$$

□

The spectral factorization for the dual case is also often used and follows by taking the transpose of the corresponding transfer matrices.

Corollary 13.22 *Assume* $G(s) := \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{RL}_\infty$ *and* $\gamma > \|G(s)\|_\infty$. *Then, there exists a transfer matrix* $M \in \mathcal{RL}_\infty$ *such that* $MM^\sim = \gamma^2 I - GG^\sim$ *and* $M^{-1} \in \mathcal{RH}_\infty$. *A particular realization of* M *is*

$$M(s) = \left[\begin{array}{c|c} A & -LR^{1/2} \\ \hline C & R^{1/2} \end{array}\right]$$

where

$$\begin{aligned} R &= \gamma^2 I - DD^* \\ L &= (YC^* + BD^*)R^{-1} \\ Y &= Ric\begin{bmatrix} (A + BD^*R^{-1}C)^* & C^*R^{-1}C \\ -B(I + D^*R^{-1}D)B^* & -(A + BD^*R^{-1}C) \end{bmatrix} \end{aligned}$$

and $Y \geq 0$ *if* A *is stable.*

For convenience, we also include the following spectral factorization results which are again special cases of Theorem 13.19.

Corollary 13.23 *Let* $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *be a stabilizable and detectable realization.*

(a) Suppose $G^\sim(j\omega)G(j\omega) > 0$ *for all* ω *or* $\begin{bmatrix} A - j\omega & B \\ C & D \end{bmatrix}$ *has full column rank for all* ω. *Let*

$$X = Ric\begin{bmatrix} A - BR^{-1}D^*C & -BR^{-1}B^* \\ -C^*(I - DR^{-1}D^*)C & -(A - BR^{-1}D^*C)^* \end{bmatrix}$$

with $R := D^*D > 0$. Then we have the following spectral factorization

$$W^\sim W = G^\sim G$$

where $W^{-1} \in \mathcal{RH}_\infty$ and

$$W = \left[\begin{array}{c|c} A & B \\ \hline R^{-1/2}(D^*C + B^*X) & R^{1/2} \end{array}\right].$$

(b) Suppose $G(j\omega)G^\sim(j\omega) > 0$ for all ω or $\begin{bmatrix} A - j\omega & B \\ C & D \end{bmatrix}$ has full row rank for all ω. Let

$$Y = Ric \begin{bmatrix} (A - BD^*\tilde{R}^{-1}C)^* & -C^*\tilde{R}^{-1}C \\ -B(I - D^*\tilde{R}^{-1}D)B^* & -(A - BD^*\tilde{R}^{-1}C) \end{bmatrix}$$

with $\tilde{R} := DD^* > 0$. Then we have the following spectral factorization

$$\tilde{W}\tilde{W}^\sim = GG^\sim$$

where $\tilde{W}^{-1} \in \mathcal{RH}_\infty$ and

$$\tilde{W} = \left[\begin{array}{c|c} A & (BD^* + YC^*)\tilde{R}^{-1/2} \\ \hline C & \tilde{R}^{1/2} \end{array}\right].$$

Theorem 13.19 also gives some additional characterizations of a transfer matrix $\mathcal{H}_\infty$ norm.

Corollary 13.24 *Let $\gamma > 0$, $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{RH}_\infty$ and*

$$H := \begin{bmatrix} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix}$$

where $R = \gamma^2 I - D^*D$. Then the following conditions are equivalent:

(i) $\|G\|_\infty < \gamma$.

(ii) $\bar{\sigma}(D) < \gamma$ *and H has no eigenvalues on the imaginary axis.*

(iii) $\bar{\sigma}(D) < \gamma$ *and $H \in dom(Ric)$.*

(iv) $\bar{\sigma}(D) < \gamma$ *and $H \in dom(Ric)$ and $Ric(H) \geq 0$ ($Ric(H) > 0$ if (C, A) is observable).*

Proof. This follows from the fact that $\|G\|_\infty < \gamma$ is equivalent to that the following function is positive definite for all ω:

$$\Phi(j\omega) := \gamma^2 I - G^T(-j\omega)G(j\omega)$$

$$= \begin{bmatrix} B^*(-j\omega I - A^*)^{-1} & I \end{bmatrix} \begin{bmatrix} -C^*C & -C^*D \\ -D^*C & \gamma^2 I - D^*D \end{bmatrix} \begin{bmatrix} (j\omega I - A)^{-1}B \\ I \end{bmatrix} > 0,$$

and the fact that

$$\begin{bmatrix} -I & 0 \\ 0 & I \end{bmatrix} H \begin{bmatrix} -I & 0 \\ 0 & I \end{bmatrix} = \begin{bmatrix} A + BR^{-1}D^*C & -BR^{-1}B^* \\ C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix}.$$

□

The equivalence between (i) and (iv) in the above corollary is usually referred as *Bounded Real Lemma*.

13.5 Positive Real Functions

A square $(m \times m)$ matrix function $G(s) \in \mathcal{RH}_\infty$ is said to be *positive real (PR)* if $G(j\omega) + G^*(j\omega) \geq 0$ for all finite ω, i.e., $\omega \in \mathbb{R}$, and $G(s)$ is said to be *strictly positive real (SPR)* if $G(j\omega) + G^*(j\omega) > 0$ for all $\omega \in \mathbb{R}$.

Theorem 13.25 *Let* $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *be a state space realization of* $G(s)$ *with* A *stable (not necessarily a minimal realization). Suppose there exist an* $X \geq 0$, Q, *and* W *such that*

$$XA + A^*X = -Q^*Q \tag{13.34}$$

$$B^*X + W^*Q = C \tag{13.35}$$

$$D + D^* = W^*W, \tag{13.36}$$

Then $G(s)$ *is positive real and*

$$G(s) + G^\sim(s) = M^\sim(s)M(s)$$

with $M(s) = \left[\begin{array}{c|c} A & B \\ \hline Q & W \end{array}\right]$. *Furthermore, if* $M(j\omega)$ *has full column rank for all* $\omega \in \mathbb{R}$, *then* $G(s)$ *is strictly positive real.*

Proof.

$$G(s)+G^{\sim}(s)=\left[\begin{array}{cc|c} A & 0 & B \\ 0 & -A^* & -C^* \\ \hline C & B^* & D+D^* \end{array}\right]=\left[\begin{array}{cc|c} A & 0 & B \\ 0 & -A^* & -(XB+Q^*W) \\ \hline B^*X+W^*Q & B^* & W^*W \end{array}\right].$$

Apply a similarity transformation $\begin{bmatrix} I & 0 \\ X & I \end{bmatrix}$ to the last realization to get

$$\begin{aligned} G(s)+G^{\sim}(s) &= \left[\begin{array}{cc|c} A & 0 & B \\ XA+A^*X & -A^* & -Q^*W \\ \hline W^*Q & B^* & W^*W \end{array}\right]=\left[\begin{array}{cc|c} A & 0 & B \\ -Q^*Q & -A^* & -Q^*W \\ \hline W^*Q & B^* & W^*W \end{array}\right] \\ &= \left[\begin{array}{c|c} -A^* & -Q^* \\ \hline B^* & W^* \end{array}\right]\left[\begin{array}{c|c} A & B \\ \hline Q & W \end{array}\right]. \end{aligned}$$

This implies that

$$G(j\omega)+G^*(j\omega)=M^*(j\omega)M(j\omega)\geq 0$$

i.e., $G(s)$ is positive real. Finally note that if $M(j\omega)$ has full column rank for all $\omega\in\mathbb{R}$, then $M(j\omega)x\neq 0$ for all $x\in\mathbb{C}^m$ and $\omega\in\mathbb{R}$. Thus $G(s)$ is strictly positive real. □

Theorem 13.26 *Suppose (A,B,C,D) is a minimal realization of $G(s)$ with A stable and $G(s)$ is positive real. Then there exist an $X\geq 0$, Q, and W such that*

$$\begin{aligned} XA+A^*X &= -Q^*Q \\ B^*X+W^*Q &= C \\ D+D^* &= W^*W. \end{aligned}$$

and

$$G(s)+G^{\sim}(s)=M^{\sim}(s)M(s)$$

with $M(s)=\left[\begin{array}{c|c} A & B \\ \hline Q & W \end{array}\right]$. Furthermore, if $G(s)$ is strictly positive real, then $M(j\omega)$ given above has full column rank for all $\omega\in\mathbb{R}$.

Proof. Since $G(s)$ is assumed to be positive real, there exists a transfer matrix $M(s)=\left[\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array}\right]$ with A_1 stable such that

$$G(s)+G^{\sim}(s)=M^{\sim}(s)M(s)$$

where A and A_1 have the same dimensions. Now let $X_1 \geq 0$ be the solution of the following Lyapunov equation:

$$X_1 A_1 + A_1^* X_1 = -C_1^* C_1.$$

Then

$$\begin{aligned}
M^\sim(s)M(s) &= \left[\begin{array}{c|c} -A_1^* & -C_1^* \\ \hline B_1^* & D_1^* \end{array}\right] \left[\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array}\right] = \left[\begin{array}{cc|c} A_1 & 0 & B_1 \\ -C_1^*C_1 & -A_1^* & -C_1^*D_1 \\ \hline D_1^*C_1 & B_1^* & D_1^*D_1 \end{array}\right] \\
&= \left[\begin{array}{cc|c} A_1 & 0 & B_1 \\ X_1A_1 + A_1^*X_1 & -A_1^* & -C_1^*D_1 \\ \hline D_1^*C_1 & B_1^* & D_1^*D_1 \end{array}\right] \\
&= \left[\begin{array}{cc|c} A_1 & 0 & B_1 \\ 0 & -A_1^* & -(X_1B_1 + C_1^*D_1) \\ \hline B_1^*X + D_1^*C_1 & B_1^* & D_1^*D_1 \end{array}\right] \\
&= D_1^*D_1 + \left[\begin{array}{c|c} A_1 & B_1 \\ \hline B_1^*X + D_1^*C_1 & 0 \end{array}\right] + \left[\begin{array}{c|c} -A_1^* & -(B_1^*X + D_1^*C_1)^* \\ \hline B_1^* & 0 \end{array}\right].
\end{aligned}$$

But the realization for $G(s) + G^\sim(s)$ is given by

$$G(s) + G^\sim(s) = D + D^* + \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array}\right] + \left[\begin{array}{c|c} -A^* & -C^* \\ \hline B^* & 0 \end{array}\right].$$

Since the realization for $G(s)$ is minimal, there exists a nonsingular matrix T such that

$$A = TA_1T^{-1}, \quad B = TB_1, \quad C = (B_1^*X + D_1^*C_1)T^{-1}, \quad D + D^* = D_1^*D_1.$$

Now the conclusion follows by defining

$$X = (T^{-1})^* X_1 T^{-1}, \quad W = D_1, \quad Q = C_1T^{-1}.$$

□

Corollary 13.27 *Let* $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *be a state space realization of* $G(s) \in \mathcal{RH}_\infty$ *with* A *stable and* $R := D + D^* > 0$. *Then* $G(s)$ *is strictly positive real if and only if there exists a stabilizing solution to the following Riccati equation:*

$$X(A - BR^{-1}C) + (A - BR^{-1}C)^*X + XBR^{-1}B^*X + C^*R^{-1}C = 0.$$

Moreover, $M(s) = \left[\begin{array}{c|c} A & B \\ \hline R^{-\frac{1}{2}}(C - B^*X) & R^{\frac{1}{2}} \end{array}\right]$ *is minimal phase and*

$$G(s) + G^\sim(s) = M^\sim(s)M(s).$$

Proof. This follows from Theorem 13.19 and from the fact that

$$G(j\omega) + G^*(j\omega) > 0$$

for all ω including ∞. □

The above corollary also leads to the following special spectral factorization.

Corollary 13.28 *Let* $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{RH}_\infty$ *with* D *full row rank and* $G(j\omega)G^*(j\omega) > 0$ *for all* ω*. Let* P *be the controllability grammian of* (A, B)*:*

$$PA^* + AP + BB^* = 0.$$

Define

$$B_W = PC^* + BD^*.$$

Then there exists an $X \geq 0$ *such that*

$$XA + A^*X + (C - B_W^*X)^*(DD^*)^{-1}(C - B_W^*X) = 0.$$

Furthermore, there is an $M(s) \in \mathcal{RH}_\infty$ *such that* $M^{-1}(s) \in \mathcal{RH}_\infty$ *and*

$$G(s)G^\sim(s) = M^\sim(s)M(s)$$

where $M(s) = \left[\begin{array}{c|c} A & B_W \\ \hline C_W & D_W \end{array}\right]$ *with a square matrix* D_W *such that*

$$D_W^*D_W = DD^*$$

and

$$C_W = D_W(DD^*)^{-1}(C - B_W^*X).$$

Proof. This corollary follows from Corollary 13.27 and the fact that $G(j\omega)G^*(j\omega) > 0$ and

$$G(s)G^\sim(s) = \left[\begin{array}{c|c} A & B_W \\ \hline C & 0 \end{array}\right] + \left[\begin{array}{c|c} -A^* & -C^* \\ \hline B_W^* & 0 \end{array}\right] + DD^*.$$

□

A dual spectral factorization can also be obtained easily.

13.6 Inner Functions

A transfer function N is called *inner* if $N \in \mathcal{RH}_\infty$ and $N^\sim N = I$ and *co-inner* if $N \in \mathcal{RH}_\infty$ and $NN^\sim = I$. Note that N need not be square. Inner and co-inner are dual notions, i.e., N is an inner iff N^T is a co-inner. A matrix function $N \in \mathcal{RL}_\infty$ is called *all-pass* if N is square and $N^\sim N = I$; clearly a square inner function is all-pass. We will focus on the characterizations of inner functions here and the properties of co-inner functions follow by duality.

Note that N inner implies that N has at least as many rows as columns. For N inner and any $q \in \mathbb{C}^m$, $v \in \mathcal{L}_2$, $\|N(j\omega)q\| = \|q\|$, $\forall\omega$ and $\|Nv\|_2 = \|v\|_2$ since $N(j\omega)^*N(j\omega) = I$ for all ω. Because of these norm preserving properties, inner matrices will play an important role in the control synthesis theory in this book. In this section, we present a state-space characterization of inner transfer functions.

Lemma 13.29 *Suppose* $N = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{RH}_\infty$ *and* $X = X^* \geq 0$ *satisfies*

$$A^*X + XA + C^*C = 0. \tag{13.37}$$

Then

(a) $D^*C + B^*X = 0$ *implies* $N^\sim N = D^*D$.

(b) (A,B) *controllable, and* $N^\sim N = D^*D$ *implies* $D^*C + B^*X = 0$.

Proof. Conjugating the states of

$$N^\sim N = \left[\begin{array}{cc|c} A & 0 & B \\ -C^*C & -A^* & -C^*D \\ \hline D^*C & B^* & D^*D \end{array}\right]$$

by $\begin{bmatrix} I & 0 \\ -X & I \end{bmatrix}$ on the left and $\begin{bmatrix} I & 0 \\ -X & I \end{bmatrix}^{-1} = \begin{bmatrix} I & 0 \\ X & I \end{bmatrix}$ on the right yields

$$\begin{aligned} N^\sim N &= \left[\begin{array}{cc|c} A & 0 & B \\ -(A^*X + XA + C^*C) & -A^* & -(XB + C^*D) \\ \hline B^*X + D^*C & B^* & D^*D \end{array}\right] \\ &= \left[\begin{array}{cc|c} A & 0 & B \\ 0 & -A^* & -(XB + C^*D) \\ \hline B^*X + D^*C & B^* & D^*D \end{array}\right]. \end{aligned}$$

Then (a) and (b) follow easily. $\square$

This lemma immediately leads to one characterization of inner matrices in terms of their state space representations. Simply add the condition that $D^*D = I$ to Lemma 13.29 to get $N^\sim N = I$.

Corollary 13.30 *Suppose* $N = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *is stable and minimal, and X is the observability grammian. Then N is inner if and only if*

(a) $D^*C + B^*X = 0$

(b) $D^*D = I$.

A transfer matrix $N_\perp$ is called a *complementary inner factor (CIF)* of N if $[N \, N_\perp]$ is square and inner. The dual notion of the complementary co-inner factor is defined in the obvious way. Given an inner N, the following lemma gives a construction of its CIF. The proof of this lemma follows from straightforward calculation and from the fact that $CX^\dagger X = C$ since $\mathrm{Im}(I - X^+X) \subset \mathrm{Ker}(X) \subset \mathrm{Ker}(C)$.

Lemma 13.31 *Let* $N = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *be an inner and X be the observability grammian. Then a CIF $N_\perp$ is given by*

$$N_\perp = \left[\begin{array}{c|c} A & -X^\dagger C^* D_\perp \\ \hline C & D_\perp \end{array}\right]$$

where $D_\perp$ is an orthogonal complement of D such that $[D \, D_\perp]$ is square and orthogonal.

13.7 Inner-Outer Factorizations

In this section, some special form of coprime factorizations will be developed. In particular, explicit realizations are given for coprime factorizations $G = NM^{-1}$ with inner numerator N and inner denominator M, respectively. The former factorization in the case of $G \in \mathcal{RH}_\infty$ will give an inner-outer factorization[1]. The results will be proven for the right coprime factorizations, while the results for left coprime factorizations follow by duality.

Let $G \in \mathcal{R}_p$ be a $p \times m$ transfer matrix and denote $R^{1/2}R^{1/2} = R$. For a given full column rank matrix D, let $D_\perp$ denote for any orthogonal complement of D so that $\begin{bmatrix} DR^{-1/2} & D_\perp \end{bmatrix}$ (with $R = D^*D > 0$) is square and orthogonal. To obtain an *rcf* of G with N inner, we note that if NM^{-1} is an rcf, then $(NZ_r)(MZ_r)^{-1}$ is also an rcf for any nonsingular real matrix Z_r. We simply need to use the formulas in Theorem 5.9 to solve for F and Z_r.

[1] A $p \times m$ ($p \leq m$) transfer matrix $G_o \in \mathcal{RH}_\infty$ is called an *outer* if $G_o(s)$ has full row rank in the open right half plane, i.e., $G_o(s)$ has full row normal rank and has no open right half plane zeros.

Theorem 13.32 *Assume $p \geq m$. Then there exists an* rcf $G = NM^{-1}$ *such that* N *is* inner *if and only if $G^{\sim}G > 0$ on the $j\omega$-axis, including at ∞. This factorization is unique up to a constant unitary multiple. Furthermore, assume that the realization of* $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *is stabilizable and that* $\begin{bmatrix} A - j\omega I & B \\ C & D \end{bmatrix}$ *has full column rank for all $\omega \in \mathbb{R}$. Then a particular realization of the desired coprime factorization is*

$$\begin{bmatrix} M \\ N \end{bmatrix} := \left[\begin{array}{c|c} A + BF & BR^{-1/2} \\ \hline F & R^{-1/2} \\ C + DF & DR^{-1/2} \end{array}\right] \in \mathcal{RH}_\infty$$

where

$$R = D^*D > 0$$

$$F = -R^{-1}(B^*X + D^*C)$$

and

$$X = Ric \begin{bmatrix} A - BR^{-1}D^*C & -BR^{-1}B^* \\ -C^*(I - DR^{-1}D^*)C & -(A - BR^{-1}D^*C)^* \end{bmatrix} \geq 0.$$

Moreover, a complementary inner factor can be obtained as

$$N_\perp = \left[\begin{array}{c|c} A + BF & -X^\dagger C^* D_\perp \\ \hline C + DF & D_\perp \end{array}\right]$$

if $p > m$.

Proof. ($\Rightarrow$) Suppose $G = NM^{-1}$ is an *rcf* and $N^{\sim}N = I$. Then $G^{\sim}G = (M^{-1})^{\sim}M^{-1} > 0$ on the $j\omega$-axis since $M \in \mathcal{RH}_\infty$.

($\Leftarrow$) This can be shown by using Corollary 13.20 first to get a factorization $G^{\sim}G = (M^{-1})^{\sim}(M^{-1})$ and then to compute $N = GM$. The following proof is more direct. It will be proven by showing that the definition of the inner and coprime factorization formula given in Theorem 5.9 lead directly to the above realization of the *rcf* of G with an inner numerator. That $G = NM^{-1}$ is an *rcf* follows immediately once it is established that F is a stabilizing state feedback. Now suppose

$$N = \left[\begin{array}{c|c} A + BF & BZ_r \\ \hline C + DF & DZ_r \end{array}\right]$$

and let F and Z_r be such that

$$(DZ_r)^*(DZ_r) = I \tag{13.38}$$

$$(BZ_r)^*X + (DZ_r)^*(C + DF) = 0 \tag{13.39}$$

$$(A+BF)^*X + X(A+BF) + (C+DF)^*(C+DF) = 0. \tag{13.40}$$

Clearly, we have that $Z_r = R^{-1/2}U$ where $R = D^*D > 0$ and where U is any orthogonal matrix. Take $U = I$ and solve (13.39) for F to get

$$F = -R^{-1}(B^*X + D^*C).$$

Then substitute F into (13.40) to get

$$\begin{aligned} 0 &= (A+BF)^*X + X(A+BF) + (C+DF)^*(C+DF) \\ &= (A - BR^{-1}D^*C)^*X + X(A - BR^{-1}D^*C) - XBR^{-1}B^*X + C^*D_\perp D_\perp^* C \end{aligned}$$

where $D_\perp D_\perp^* = I - DR^{-1}D^*$. To show that such choices indeed make sense, we need to show that $H \in dom(Ric)$, where

$$H = \begin{bmatrix} A - BR^{-1}D^*C & -BR^{-1}B^* \\ -C^*D_\perp D_\perp^* C & -(A - BR^{-1}D^*C)^* \end{bmatrix}$$

so $X = Ric(H)$. However, by Theorem 13.19, $H \in dom(Ric)$ is guaranteed by the fact that $\begin{bmatrix} A - j\omega & B \\ C & D \end{bmatrix}$ has full column rank (or $G^\sim(j\omega)G(j\omega) > 0$.

The uniqueness of the factorization follows from coprimeness and N inner. Suppose that $G = N_1M_1^{-1} = N_2M_2^{-1}$ are two right coprime factorizations and that both numerators are inner. By coprimeness, these two factorizations are unique up to a right multiple which is a unit[2] in $\mathcal{RH}_\infty$. That is, there exists a unit $\Theta \in \mathcal{RH}_\infty$ such that $\begin{bmatrix} M_1 \\ N_1 \end{bmatrix}\Theta = \begin{bmatrix} M_2 \\ N_2 \end{bmatrix}$. Clearly, Θ is inner since $\Theta^\sim\Theta = \Theta^\sim N_1^\sim N_1 \Theta = N_2^\sim N_2 = I$. The only inner units in $\mathcal{RH}_\infty$ are constant matrices, and thus the desired uniqueness property is established. Note that the non-uniqueness is contained entirely in the choice of a particular square root of R.

Finally, the formula for $N_\perp$ follows from Lemma 13.31. □

Note that the important inner-outer factorization formula can be obtained from this inner numerator coprime factorization if $G \in \mathcal{RH}_\infty$.

Corollary 13.33 *Suppose $G \in \mathcal{RH}_\infty$; then the denominator matrix M in Theorem 13.32 is an outer. Hence, the factorization $G = N(M^{-1})$ given in Theorem 13.32 is an inner-outer factorization.*

Remark 13.7 It is noted that the above inner-outer factorization procedure does not apply to the strictly proper transfer matrix even if the factorization exists. For example, $G(s) = \frac{s-1}{s+1}\frac{1}{s+2}$ has inner-outer factorizations but the above procedure cannot be used. The inner-outer factorization for the general transfer matrices can be done using the method adopted in Section 6.1 of Chapter 6. ♡

[2] A function Θ is called a unit in $\mathcal{RH}_\infty$ if $\Theta, \Theta^{-1} \in \mathcal{RH}_\infty$.

Suppose that the system G is not stable; then a coprime factorization with an inner denominator can also be obtained by solving a special Riccati equation. The proof of this result is similar to the inner numerator case and is omitted.

Theorem 13.34 *Assume that* $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{R}_p$ *and* (A, B) *is stabilizable. Then there exists a right coprime factorization* $G = NM^{-1}$ *such that* $M \in \mathcal{RH}_\infty$ *is inner if and only if* G *has no poles on* $j\omega$*-axis. A particular realization is*

$$\begin{bmatrix} M \\ N \end{bmatrix} := \left[\begin{array}{c|c} A+BF & B \\ \hline F & I \\ C+DF & D \end{array}\right] \in \mathcal{RH}_\infty$$

where

$$F = -B^*X$$
$$X = Ric \begin{bmatrix} A & -BB^* \\ 0 & -A^* \end{bmatrix} \geq 0.$$

Dual results can be obtained when $p \leq m$ by taking the transpose of the transfer function matrix. In these factorizations, output injection using the dual Riccati solution replaces state feedback to obtain the corresponding left factorizations.

Theorem 13.35 *Assume* $p \leq m$*. Then there exists an* lcf $G = \tilde{M}^{-1}\tilde{N}$ *such that* $\tilde{N}$ *is a* co-inner *if and only if* $GG^{\sim} > 0$ *on the* $j\omega$*-axis, including at* ∞*. This factorization is unique up to a constant unitary multiple. Furthermore, assume that the realization of* $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *is detectable and that* $\begin{bmatrix} A - j\omega I & B \\ C & D \end{bmatrix}$ *has full row rank for all* $\omega \in \mathbb{R}$*. Then a particular realization of the desired coprime factorization is*

$$\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} := \left[\begin{array}{c|cc} A+LC & L & B+LD \\ \hline \tilde{R}^{-1/2}C & \tilde{R}^{-1/2} & \tilde{R}^{-1/2}D \end{array}\right] \in \mathcal{RH}_\infty$$

where

$$\tilde{R} = DD^* > 0$$
$$L = -(BD^* + YC^*)\tilde{R}^{-1}$$

and

$$Y = Ric \begin{bmatrix} (A - BD^*\tilde{R}^{-1}\tilde{C})^* & -C^*\tilde{R}^{-1}C \\ -B(I - D^*\tilde{R}^{-1}D)B & -(A - BD^*\tilde{R}^{-1}C) \end{bmatrix} \geq 0.$$

Moreover, a complementary co-inner factor can be obtained as

$$\tilde{N}_\perp = \left[\begin{array}{c|c} A+LC & B+LD \\ \hline -\tilde{D}_\perp B^* Y^\dagger & \tilde{D}_\perp \end{array}\right]$$

if $p < m$*, where* $\tilde{D}_\perp$ *is a full row rank matrix such that* $\tilde{D}_\perp^*\tilde{D}_\perp = I - D^*\tilde{R}^{-1}D$*.*

Theorem 13.36 *Assume that* $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{R}_p$ *and* (C, A) *is detectable. Then there exists a left coprime factorization* $G = \tilde{M}^{-1}\tilde{N}$ *such that* $\tilde{M} \in \mathcal{RH}_\infty$ *is inner if and only if* G *has no poles on* $j\omega$*-axis. A particular realization is*

$$\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} := \left[\begin{array}{c|cc} A + LC & L & B + LD \\ \hline C & I & D \end{array}\right] \in \mathcal{RH}_\infty$$

where

$$L = -YC^*$$

$$Y = Ric \begin{bmatrix} A^* & -C^*C \\ 0 & -A \end{bmatrix} \geq 0.$$

13.8 Normalized Coprime Factorizations

A right coprime factorization of $G = NM^{-1}$ with $N, M \in \mathcal{RH}_\infty$ is called a *normalized right coprime factorization* if

$$M^\sim M + N^\sim N = I$$

i.e., if $\begin{bmatrix} M \\ N \end{bmatrix}$ is an inner. Similarly, an *lcf* $G = \tilde{M}^{-1}\tilde{N}$ is called a *normalized left coprime factorization* if $\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}$ is a co-inner.

The normalized coprime factorization is easy to obtain from the definition. The following theorem can be proven using the same procedure as in the proof for the coprime factorization with inner numerator. In this case, the proof involves choosing F and Z_r such that $\begin{bmatrix} M \\ N \end{bmatrix}$ is an inner.

Theorem 13.37 *Let a realization of* G *be given by*

$$G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

and define

$$R = I + D^*D > 0, \qquad \tilde{R} = I + DD^* > 0.$$

(a) *Suppose* (A, B) *is stabilizable and* (C, A) *has no unobservable modes on the imaginary axis. Then there is a normalized right coprime factorization* $G = NM^{-1}$

$$\begin{bmatrix} M \\ N \end{bmatrix} := \left[\begin{array}{c|c} A + BF & BR^{-1/2} \\ \hline F & R^{-1/2} \\ C + DF & DR^{-1/2} \end{array}\right] \in \mathcal{RH}_\infty$$

where

$$F = -R^{-1}(B^*X + D^*C)$$

and

$$X = Ric \begin{bmatrix} A - BR^{-1}D^*C & -BR^{-1}B^* \\ -C^*\tilde{R}^{-1}C & -(A - BR^{-1}D^*C)^* \end{bmatrix} \geq 0.$$

(b) *Suppose (C, A) is detectable and (A, B) has no uncontrollable modes on the imaginary axis. Then there is a normalized left coprime factorization $G = \tilde{M}^{-1}\tilde{N}$*

$$\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} := \left[\begin{array}{c|cc} A + LC & L & B + LD \\ \hline \tilde{R}^{-1/2}C & \tilde{R}^{-1/2} & \tilde{R}^{-1/2}D \end{array}\right]$$

where

$$L = -(BD^* + YC^*)\tilde{R}^{-1}$$

and

$$Y = Ric \begin{bmatrix} (A - BD^*\tilde{R}^{-1}C)^* & -C^*\tilde{R}^{-1}C \\ -BR^{-1}B^* & -(A - BD^*\tilde{R}^{-1}C) \end{bmatrix} \geq 0.$$

(c) *The controllability grammian P and observability grammian Q of $\begin{bmatrix} M \\ N \end{bmatrix}$ are given by*

$$P = (I + YX)^{-1}Y, \quad Q = X$$

while the controllability grammian $\tilde{P}$ and observability grammian $\tilde{Q}$ of $\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}$ are given by

$$\tilde{P} = Y, \quad \tilde{Q} = (I + XY)^{-1}X.$$

Proof. We shall only prove the first part of (c). It is obvious that $Q = X$ since the Riccati equation for X can be written as

$$X(A + BF) + (A + BF)^*X + \begin{bmatrix} F \\ C + DF \end{bmatrix}^* \begin{bmatrix} F \\ C + DF \end{bmatrix} = 0$$

while the controllability grammian solves the Lyapunov equation

$$(A + BF)P + P(A + BF)^* + BR^{-1}B^* = 0$$

or equivalently

$$P = \text{Ric} \begin{bmatrix} (A + BF)^* & 0 \\ -BR^{-1}B^* & -(A + BF) \end{bmatrix}.$$

Now let $T = \begin{bmatrix} I & X \\ 0 & I \end{bmatrix}$; then

$$\begin{bmatrix} (A+BF)^* & 0 \\ -BR^{-1}B^* & -(A+BF) \end{bmatrix} = T \begin{bmatrix} A - BD^*\tilde{R}^{-1}C & -C^*\tilde{R}^{-1}C \\ -BR^{-1}B^* & -(A - BD^*\tilde{R}^{-1}C)^* \end{bmatrix} T^{-1}.$$

This shows that the stable invariant subspaces for these two Hamiltonian matrices are related by

$$\mathcal{X}_- \begin{bmatrix} (A+BF)^* & 0 \\ -BR^{-1}B^* & -(A+BF) \end{bmatrix} = T\ \mathcal{X}_- \begin{bmatrix} A - BD^*\tilde{R}^{-1}C & -C^*\tilde{R}^{-1}C \\ -BR^{-1}B^* & -(A - BD^*\tilde{R}^{-1}C)^* \end{bmatrix}$$

or

$$\operatorname{Im} \begin{bmatrix} I \\ P \end{bmatrix} = T \operatorname{Im} \begin{bmatrix} I \\ Y \end{bmatrix} = \operatorname{Im} \begin{bmatrix} I + XY \\ Y \end{bmatrix} = \operatorname{Im} \begin{bmatrix} I \\ Y(I+XY)^{-1} \end{bmatrix}.$$

Hence we have $P = Y(I+XY)^{-1}$. □

13.9 Notes and References

The general solutions of a Riccati equation are given by Martensson [1971]. The iterative procedure for solving ARE was first introduced by Kleinman [1968] for a special case and was further developed by Wonham [1968]. It was used by Coppel [1974], Ran and Vreugdenhil [1988], and many others for the proof of the existence of maximal and minimal solutions. The comparative results were obtained in Ran and Vreugdenhil [1988]. The paper by Wimmer[1985] also contains comparative results for some special cases. The paper by Willems [1971] contains a comprehensive treatment of ARE and the related optimization problems. Some matrix factorization results are given in Doyle [1984]. Numerical methods for solving ARE can be found in Arnold and Laub [1984], Dooren [1981], and references therein. The state space spectral factorization for functions singular at ∞ or on imaginary axis is considered in Clements and Glover [1989] and Clements [1993].